

Protocol

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Balloon Angioplasty vs. Aggressive Medical Management for Symptomatic Intracranial Artery Stenosis: The BASIS Randomized Clinical Trial

Versions of BASIS Protocols

BASIS Protocol V2.0-20180625 (original version)

BASIS Protocol V2.1-20200414 (amended version)

BASIS Protocol V2.2-20211222 (amended version)

BASIS Protocol V2.3-20220328 (final version)

This supplement contains the following items:**1. Original protocol in English (page 3 to 59), final protocol in English (page 60 to 123)****2. Original statistical analysis plan in English (page 124 to 139), final statistical analysis plan in English (page 140 to 158)**

**Balloon Angioplasty vs. Aggressive Medical Management for
Symptomatic Intracranial Artery Stenosis: The BASIS Randomized
Clinical Trial
(BASIS)**



Protocol

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Protocol Version 2.0

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Protocol Signature Page

I have read this protocol and agree to adhere to the requirements.

By signing this document we confirm that the clinical study will be conducted in accordance with the protocol and all applicable laws and regulations including, but not limited to, the International Conference on Harmonisation Guideline for Good Clinical Practice (GCP) and the ethical principles that have their origins in the Declaration of Helsinki.

Clinical Site

Site Principal Investigator Signature

Date

1. BASIS Protocol Synopsis

Official Title	Balloon Angioplasty for Symptomatic Intracranial Artery Stenosis: A Multicenter, Prospective, Open-Label, Blinded-Endpoint, Randomized, Controlled Trial
National Subject and Number	Prediction model and intervention strategy for recurrence and progression of symptomatic intracranial and extracranial major artery stenosis (2017YFC1307900)
Acronym	BASIS
PI	Prof. Zhongrong Miao and Prof. Yilong Wang
Study Objectives	To evaluate whether combined balloon angioplasty and aggressive medical management (AMM) is superior to AMM for patients with high-risk intracranial atherosclerotic stenosis (ICAS)
Study Design	The BASIS trial is an investigator-initiated, multicenter, prospective, randomized, open-label, blinded end-point (PROBE) trial Patients eligible in BASIS were 35 to 80 years old, with a recent transient ischemic attack (<90 days) or ischemic stroke (21 to 90 days) before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial artery (anterior/posterior circulation). Patients were randomly assigned to the experimental group and control group Experimental group: patients will receive balloon angioplasty plus AMM Control group: patients will receive the AMM alone ✓ AMM : aspirin 100 mg daily for the entire follow-up and clopidogrel 75mg daily for the first 90 days after enrollment. Whether to continue taking clopidogrel after 90 days will be assessed according to the patient's condition. Clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance. Atorvastatin will be taken

	at 20mg-80mg/day for one year after enrollment; the dose will be adjusted to target LDL <1.8mmol/L or <70mg/dl. The decision of whether to continue taking atorvastatin after one year will be assessed according to the patient's condition. ✓ Vascular risk factor management— blood pressure target 130-140/90-100 mmHg
Primary Endpoint	The primary outcome is stroke or death within 30 days after enrollment, or any ischemic stroke or revascularization of the qualifying artery after 30 days through 12 months after enrollment.
Secondary Endpoints	- Any stroke (ischemic or haemorrhagic stroke) in the territory of the target artery or all-cause death within 90 days after enrollment - Any stroke (ischemic or haemorrhagic stroke) outside the territory of the target artery within 90 days after enrollment - mRS (modified Rankin Scale) at 90 days - Any stroke (ischemic or haemorrhagic stroke) in the territory of the target artery or all-cause death within 12 months after enrollment - Any stroke (ischemic or haemorrhagic stroke) outside the territory of the target artery within 12 months after enrollment - Revascularization of the qualifying artery within 12 months after enrollment - mRS at 12 months - Restenosis rate of the target artery within 12 months (defined as rate of stenosis >70% or increased by 30% based on follow-up neurovascular imaging) - Combined events such as stroke, myocardial infarction, and vascular death within 12 months after enrollment - Quality of life assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire) within 12 months after enrollment.

Inclusion Criteria	1. Age 35-80 years 2. Participants with primary or recurrent sICAS (a recent TIA[<90 days] or ischemic stroke [21-90 days] before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial artery receiving treatment with at least one antithrombotic drug and/or standard medical management of vascular risk factors 3. Diagnosed by DSA: a major intracranial artery (anterior/posterior circulation) severe atherosclerotic stenosis (70-99% according to WASID method) with lesion length $\leq$ 10mm, diameter $\geq$ 1.5mm, and normal distal artery. (Regarding the curvature and angle of the lesion, whether the patient is enrolled in BASIS is per the investigator's determination based on the patient's anatomy); 4. Informed consent signed.
Exclusion Criteria	1. Surgery within the past 30 days or plan to receive major surgery within 90 days 2. Thrombolytic therapy within 24 hours before enrollment 3. Neurological deficit worsening within 24 hours before enrollment 4. Acute ischemic stroke onset within 21 days before enrollment 5. Other intracranial arteries with severe stenosis (70-99%) besides the target artery and its supplying artery 6. Target artery's parent artery stenosis $> 50\%$. For example, patient with MCA severe stenosis (target artery) and ipsilateral ICA stenosis $> 50\%$ should be excluded; patient with BA severe stenosis (target artery) and dominant VA stenosis $> 50\%$ should be excluded; the patient with extracranial artery (non-lesion side) stenosis $> 70\%$ should be excluded 7. Baseline mRS ≥ 3 8. Non-atherosclerotic disease (e.g., arterial dissection, moyamoya

	disease, vascular inflammatory lesions caused by infection, autoimmune disease, post-radiation, postpartum status; developmental or genetic abnormalities such as fibromuscular dysplasia, sickle cell anemia, suspected vasospasm) 9. Target artery with severely calcified stenosis 10. Suspected ischemic event caused by embolism or the arterial embolism originates from an extracranial segment (ipsilateral chest or neck vascular steno-occlusive disease) or potential cardioembolism (e.g., atrial fibrillation, mitral stenosis, patent foramen ovale, left ventricular thrombus, myocardial infarction within 6 weeks, etc.) 11. Coexistent intracranial tumor, aneurysm or arteriovenous malformation 12. Intracranial hemorrhage within the past 3 months, including parenchymal hemorrhage, intraventricular hemorrhage, subarachnoid hemorrhage, epidural hemorrhage or subdural hemorrhage, etc. 13. Angioplasty procedure (including balloon dilatation, stenting, or endarterectomy) already performed at the original target vessel or its primary supplying artery, or a planned stenting procedure; 14. Unable to receive dual antiplatelet therapy due to other disease 15. Tortuous vascular approach that cannot be stabilized to obtain vascular access 16. Allergy to heparin, aspirin, clopidogrel, contrast agents, anesthetics, and balloon components 17. Haemoglobin < 100g/L, platelet count < $100 \times 10^9/L$, international normalized ratio (INR) >1.5 (irreversible), coagulation dysfunction or uncorrectable bleeding factors 18. Severe liver or kidney dysfunction. ALT > three-fold higher than the upper limit of normal value or AST > three-fold higher than the upper limit of normal value, or serum creatinine > two-fold higher than the upper limit of normal value
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	19. Pregnant and lactating women 20. Participants with renal artery, radial artery, or coronary artery requiring concurrent intervention 21. Expected survival is less than 1 year 22. Unable to complete follow-up due to mental illness, cognitive or emotional disorders 23. Participant in another drug/medical device clinical trial who had not yet completed the study requirements 24. Participant not suitable for the BASIS trial per investigator's opinion.
Sample Size	The estimation of the sample size is based on the assumption that the time when an event (stroke or death) occurs follows an exponential distribution. According to the 1-year results of the SAMMPRIS trial, the incidence of target vessel stroke or death was 12.2% within 12 months after enrollment in the medical management group. According to the SAMMPRIS study, the incidence of the primary endpoint events was assumed to be 6.25% within 12 months after balloon angioplasty for intracranial atherosclerosis. In this way, the sample risk rates of the experimental group and the control group were estimated to be 6.45% and 13.01%, respectively. It was assumed that 5% of patients in the control group needed balloon dilatation after 3 months of follow-up. The risk of balloon dilation in the experimental group was: $\lambda_{mcb} = 0.25\lambda_m + 0.75\lambda_b = 8.09\%$. Therefore, the adjusted risk rate of the control group was determined as: $\lambda_m^* = 0.05\lambda_{mcb} + 0.95\lambda_m = 12.76\%$. The test of the null hypothesis applies a two-sided log-rank test. SAS 9.3 or EAST 6.0 software was used to estimate sample size. The sample size was 686 cases, and a total of 91 endpoint events were estimated in the two groups, with 90% confidence, bilateral $\alpha = 0.05$, to detect 50 % relative risk reduction. It is assumed that the Lan-DeMets loss function will lead to 6% expansion of the total type I error. The SAS software is used to calculate the sample size required

	for the mid-term analysis, which increases the sample size to the required 728 cases. Considering the 10% attrition rate and random ratio, the final sample size of this study is 802 cases.
Follow-up	All participants will be followed up by the on-site neurologist at baseline, the day of DSA, discharge, 30 ± 7 days, 90 ± 7 days, 6 months ± 14 days, 1year ± 30 days and up to 3years (at 6 months intervals after 1 year).
Study introduction period	After the study was initiated at a site, the participating study centers copied the video (including the DSA and endovascular process) of 3-5 cases of balloon angioplasty in their centers and transferred the video to the core laboratory. After a review was passed, the group continued enrollment, to ensure consistency of the balloon angioplasty process. The first 100 participants completed the peri-operative (30 days) observation, mainly collecting adverse events during the peri-operative period. The PI met with other investigators from the participating centers for interim discussion to evaluate the feasibility of this treatment plan and the continuation of later research.

2. Abbreviations

ACA	Anterior Cerebral Artery
ACE	Angiotensin Converting Enzyme
ADA	American Diabetes Association
AE	Adverse Event
ARB	Angiotensin Receptor Blocker
BMI	Body Mass Index
BP	Blood Pressure
CEC	Clinical Events Adjudication Committee
CFDA	China Food and Drug Administration
CHANCE	Clopidogrel in High-risk patients with Acute Nondisabling Cerebrovascular Events
CI	Confidence Interval
CRF	Case Report Form
CT	Computer Tomography
CTP	Computed Tomography Perfusion Imaging
DAPT	Dual Antiplatelet Therapy
DICOM	Digital Imaging and Communications in Medicine
DSA	Digital Subtraction Angiography
DSMB	Data Safety Monitoring Board
EoT	End of Trial
EQ-5D	EuroQOL five Dimensions questionnaire
FAS	Full Analysis Set
FFR	Fractional Flow Reserve
HDL	High Density Lipoprotein
ICAD	Intracranial Atherosclerotic Disease
ICAS	Intracranial Atherosclerotic Stenosis
IDL	Intermediate Density Lipoproteins
IEC	Independent Ethics Committee

IRB	Institutional Review Board
IVRS	Inter-voice Response System Inter-voice
LDL	Low Density Lipoprotein
LFT	Liver Function Test
MCA	Middle Cerebral Artery
MI	Myocardial Infarction
MoCA	Cognitive Functional Outcome by the Montreal Cognitive Assessment
mRS	Modified Rankin Scale
NMPA	National Medical Products Administration
PTAS	Percutaneous Transluminal Angioplasty and Stenting
SAE	Serious Adverse Event
SAMMPRIS	Stenting and Aggressive Medical Management for Preventing Recurrent stroke in Intracranial Stenosis
SAP	Statistical Analysis Plan
SAS	Statistical Analysis Software SAS
SNPs	Single Nucleotide Polymorphisms
TIA	Transient Ischemic Attacks
UADE	Unanticipated Adverse Device Effect
VLDL	Very Low Density Lipoprotein
WASID	Warfarin-Aspirin Symptomatic Intracranial Disease
TCD	Transcranial Doppler

3. Schedule of assessments

Visit	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5	Visit 6 Telephone interview	Visit 7	Follow up 1y to 3y (Visit 8,9,10,11)
Assessment	Baseline 15d~0 d	Angiogram 0 d	Discharge	30d±7d	90d±7 d	6 mo±14 d	1y±30 d	6 months intervals
Informed consent	X							
Inclusion/exclusion criteria check	X							
History and physical examination	X							
Vital signs ¹	X	X	X	X	X		X	
Blood laboratory test ² and urine routine test	X		X	X	X		X	
Coagulation function ³	X							
FBG, HbA1c ⁴	X			X	X		X	
Platelet aggregation test and/or CYPC219 genotype test ⁵	X							
ECG	X							
MRI/CT	X						X	
CTP ⁶	X ⁹				X		X	
DSA/CTA/MRA	X ⁹						X	
Life-style modification review ⁷	X		X	X	X	X	X	X
mRS	X			X	X	X	X	X
NIHSS (neurological examination)	X		X		X		X	
EQ-5D	X						X	X
MoCA ⁵	X						X	X
HR-VWI,TCD ⁵	X						X	
Medication review and patient compliance survey		X	X	X	X		X	
AE、SAE and endpoints ⁸	X	X	X	X	X	X	X	X ⁸

1. Vital signs: blood pressure, heart rate, respiration rate, body temperature;

2. Blood laboratory test includes blood routine and biochemistry examinations, hepatic and renal function tests.

3. Coagulation function: PT, APTT, TT, FIB, INR;

4. HbA1c is an optional test, but is recommended for patients with Diabetes Mellitus.

5. Platelet aggregation test, CYPC219 genotype test, MoCA, HR-VWI and TCD test are optional according to the center;

6. CTP is optional, but is recommended for patients with hemodynamic compromise, poor collateral or perforator stroke.

7. Life-style modification review: including whether to smoke and the number of cigarettes per day; Weight: reflect the patient's weight control through the patient's weight and BMI; Exercise: number of exercises per week and duration of each exercise.

8. After one year of follow-up, follow-up will focus on AE, SAE and endpoint events;

Abbreviations: FBG, fasting blood glucose; HbA1c, Glycosylated Hemoglobin; ECG, electrocardiograph; MRI, magnetic resonance imaging; CT, computerized tomography; CTP, CT perfusion; DSA, digital subtraction angiography; CTA, computerized tomography angiography; MRA, magnetic resonance angiography; digital subtraction angiography (DSA), computerized tomography angiography (CTA), or magnetic resonance angiography (MRA); mRS, modified Rankin scale; EQ-5D, quality-of-life EuroQol-5

177 Dimensions; MoCA, Montreal cognitive assessment; HR-VWI, High-resolution vessel wall imaging; TCD,
178 transcranial Doppler; AE, adverse event; SAE, serious adverse event.

4. Background

Intracranial atherosclerotic stenosis (ICAS) is a major etiology of stroke worldwide, which is associated with a high rate of stroke recurrence, substantial morbidity, and mortality, accounting for up to 50% of ischemic stroke in south and east Asia.¹ Despite receiving treatment with aspirin and standard medical management of vascular risk factors, patients with symptomatic ICAS (sICAS) still had as high as 23% stroke recurrence rate at 1 year.^{2,3} Therefore, optimal secondary stroke prevention and reducing mortality risk in patients with sICAS is a major global health problem. Currently, treatment options for sICAS include aggressive medical management (AMM), balloon angioplasty, and stenting (balloon angioplasty combined with stenting); however, the optimal treatment for sICAS remains unclear.

The SAMMPRIS and VISSIT trials showed high perioperative complication rates and no significant advantage of intracranial stenting compared to aggressive medical therapy (AMM) in the long term.^{4, 5} The CASSISS trial reported that stenting by experienced operators didn't add additional risks of stroke or death within 30 days compared to AMM for sICAS, although it failed to demonstrate a benefit of stenting.⁶ As lower rates of short-term stroke or mortality (peri-procedural or mean follow-up ≤ 3 months) were found in balloon angioplasty compared to intracranial stenting,⁷ balloon angioplasty alone may be a feasible alternative to stenting for sICAS treatment. Moreover, several meta-analyses also proved the safety and efficacy of balloon angioplasty for sICAS, which suggested that balloon angioplasty alone may be a potentially promising treatment for sICAS.⁸⁻¹⁰

To date, limited evidence is available from RCTs to determine the efficacy and safety of balloon angioplasty for sICAS patients. Hence, we designed the BASIS (Balloon Angioplasty for Symptomatic Intracranial Artery Stenosis) randomized controlled trial to explore whether balloon angioplasty plus AMM is superior to AMM alone in patients with sICAS.

5. Objectives and Hypothesis

5.1 Objectives

To investigate whether balloon angioplasty combined with AMM is superior to AMM alone for patients with high-risk sICAS in the primary endpoint (ischemic/hemorrhage stroke or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization from the qualifying artery beyond 30 days through 12 months after enrollment).

5.2 Hypothesis

Compared to AMM alone, balloon angioplasty combined with AMM could decrease by 50% the risk of the primary endpoint in patients with high-risk sICAS over a 12-month follow-up period.

6. Study Design

Study introduction period

After the study is formally initiated at a site, the participating study center will copy the video (including the DSA and endovascular process) of 3 to 5 cases of balloon angioplasty in their center and transfer the video to the core laboratory (Beijing Tiantan Hospital). After the review is passed, the group can continue enrollment, so as to ensure the consistency of the process of balloon angioplasty. The annual volume of BA by the neurointerventionalists in the BASIS trial was more than 50 cases. The neurointerventionalists from the participating centers received standard training in technique from the leading center at regular intervals.

The BASIS trial is an investigator-initiated, multicenter, prospective, randomized, open-label, blinded end-point (PROBE) trial that plans to enroll patients with sICAS (defined as a recent TIA (<90 days) or ischemic stroke (21-90 days) attributed to a 70-99% atherosclerotic stenosis of a major intracranial artery (anterior/posterior circulation). The patients are randomly assigned to either the experimental group or the control group at a 1:1 ratio.

228 **Experimental group:** patients will receive balloon angioplasty (details in procedure part) plus
229 AMM (same as the control group);

230 **Control group:** patients will receive AMM alone.

231 ✓ Drug administration plan of AMM: aspirin 100 mg daily for the entire follow-up and
232 clopidogrel 75mg daily for the first 90 days after enrollment. Whether to continue taking
233 clopidogrel after 90 days will be assessed according to the condition of the patient.
234 Clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients
235 with clopidogrel resistance. Atorvastatin tablets will be used 20mg-80mg/day for one year
236 after enrollment; the dose will be adjusted to control LDL <1.8mmol/L or <70mg/dl. Whether
237 to continue taking atorvastatin after one year will be assessed according to the patient's
238 condition.

239 ✓ Risk factor management—Control blood pressure 130-140/90-100 mmHg and LDL
240 <1.8mmol/L or <70mg/dl

241 The first 100 participants will complete the peri-operative (30 days) observation, mainly to collect
242 adverse events during the peri-operative period. The PI will meet with other investigators from the
243 participating centers for interim discussion to evaluate the feasibility of this treatment plan and the
244 continuation of later research, and follow-up research can be conducted later.

245

246 **7. Study Endpoint**

247 **7.1 Primary endpoint**

248 The primary endpoint is stroke or death within 30 days after enrollment, or any ischemic stroke or
249 revascularization of the qualifying artery beyond 30 days through 12 months after enrollment.

250 Ischemic stroke is defined as a new focal neurologic deficit of sudden onset, which is confirmed
251 on brain CT or MRI. Symptomatic brain hemorrhage is defined as parenchymal, subarachnoid, or
252 intraventricular hemorrhage detected on CT or MRI, which is associated with new neurologic
253 symptoms (headache, change in the level of consciousness, or focal neurologic symptoms), lasting

over 24 hours or a seizure. Symptomatic brain hemorrhage will be considered a primary endpoint only when it occurs within 30 days after enrollment or within 30 days after the balloon angioplasty during follow-up. Revascularization of the culprit artery will be considered a primary endpoint if it occurs from 30 days through 12 months after enrollment and fulfills one of the following requirements:

- a. Acute revascularization: acute qualifying artery occlusion accompanied by neurological deficit, requiring intravenous thrombolysis, intraarterial thrombolysis, mechanical thrombectomy, balloon/stent angioplasty, or intracranial-extracranial bypass.
- b. Elective revascularization: neurologic symptom-driven revascularization, including BA, stent implantation, or intracranial-extracranial bypass, if the participant fulfills one of the following conditions:
 - 1) Ischemic stroke caused by the culprit artery stenosis: a new focal neurological deficit of sudden onset attributed to the territory of the culprit artery, which is confirmed as a recurrent stroke on brain CT or MRI (follow-up imaging will be compared to baseline imaging for the detection of new lesions);
 - 2) Culprit artery stenosis that causes recurrent transient ischemic attack lasting longer than 10 minutes or new disabling neurological symptoms (paroxysmal limb weakness/numbness, inarticulateness, diplopia, or ataxia) compared to the baseline qualifying symptoms.

7.2 Secondary endpoints

1. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 90 days after enrollment
2. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 90 days after enrollment
3. mRS at 90 days

- 278 4. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death
279 within 12 months after enrollment
- 280 5. Any stroke (ischemic or hemorrhage stroke) outside the territory of the target artery within 12
281 months after enrollment
- 282 6. Revascularization of the qualifying artery within 12 months after enrollment
- 283 7. mRS at 12 months
- 284 8. Restenosis of the target artery within 12 months (defined as stenosis >70% or increased by 30%
285 based on following-up neurovascular imaging)
- 286 9. Combined events such as stroke, myocardial infarction, and vascular death within 12 months
287 after enrollment
- 288 10. Quality of life assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire) within 12
289 months after enrollment.

290 **8. Inclusion and Exclusion Criteria**

291 **8.1 Inclusion criteria**

- 292 1. Age 35-80 years
- 293 2. Participants with primary or recurrent sICAS (a recent TIA [<90 days] or ischemic stroke [14-90
294 days] before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial
295 artery) receiving treatment with at least one antithrombotic drug and/or standard medical
296 management of vascular risk factors
- 297 3. Diagnosed by DSA: a major intracranial artery (terminal internal carotid artery [ICA] [C4-C7
298 segments], middle cerebral artery [MCA] M1 segment, vertebral artery [VA] V4 segment, and
299 basilar artery [BA]) severe atherosclerotic stenosis (70-99% according to WASID method¹¹)
300 with lesion length ≤ 10 mm, diameter ≥ 1.5 mm, and normal distal artery. (Regarding the curvature

and angle of the lesion, whether the patient is enrolled in BASIS will be according to the investigator's determination based on the patient's anatomy

4. Informed consent signed

8.2 Exclusion criteria

1. Surgery within the past 30 days or plan to receive ≥ 3 -grade surgery within 90 days
2. Thrombolytic therapy within 24 hours before enrollment
3. Neurological deficit worsening within 24 hours before enrollment
4. Acute ischemic stroke onset within 21 days before enrollment
5. Other intracranial arteries with severe stenosis (70-99%) beside the target artery and its supplying artery
6. Target artery parent artery stenosis $> 50\%$. For example, patient with MCA severe stenosis (target artery) and ipsilateral ICA stenosis $> 50\%$ should be excluded; patient with BA severe stenosis (target artery) and dominant VA stenosis $> 50\%$ should be excluded; the patient with extracranial artery (non-lesion side) stenosis $> 70\%$ should be excluded
7. Baseline modified Rankin Scale ≥ 3
8. Non-atherosclerotic diseases (e.g., arterial dissection, moyamoya disease, vascular inflammatory lesions caused by infection, autoimmune diseases, post-radiation, postpartum status; developmental or genetic abnormalities such as fibromuscular dysplasia, sickle cell anemia, suspected vasospasm)
9. Target artery with severely calcified stenosis
10. Suspected ischemic event caused by embolism or the arterial embolism from extracranial segment (ipsilateral chest or neck vascular occlusive disease) or potential cardioembolism (e.g., atrial fibrillation, mitral stenosis, patent foramen ovale, left ventricular thrombus, myocardial infarction within 6 weeks, etc.)

11. Coexistent intracranial tumor, aneurysm or arteriovenous malformation
12. Intracranial hemorrhage within the past 3 months, including parenchymal hemorrhage, intraventricular hemorrhage, subarachnoid hemorrhage, epidural hemorrhage or subdural hemorrhage, etc.
13. Angioplasty procedure (including balloon dilatation, stenting, or endarterectomy) already performed at the original target vessel or its primary supplying artery, or a planned stenting procedure
14. Unable to receive dual antiplatelet therapy due to other diseases
15. Tortuous vascular approach not amenable to vascular access
16. Allergy to heparin, aspirin, clopidogrel, contrast agents, anesthetics, and balloon components
17. Hemoglobin < 100g/L, platelet count < 100×10^9 /L, international normalized ratio (INR) >1.5 (irreversible), coagulation dysfunction or uncorrectable bleeding factors
18. Severe liver or kidney dysfunction. ALT > three-fold higher than the upper limit of normal value or AST > three-fold higher than the upper limit of normal value, or serum creatinine > two-fold higher than the upper limit of normal value
19. Pregnant and lactating women
20. Participants with renal artery, radial artery, or coronary artery requiring concurrent intervention
21. Expected survival is less than 1 year
22. Unable to complete follow-up due to mental illness, cognitive or emotional disorder
23. Participants participating in other drug/medical device clinical trials who have not yet completed the study requirement
24. Participants are not suitable for the BASIS trial per investigator's opinion.

9. Informed Consent

The clinical trial must follow the Helsinki Declaration and the relevant clinical trial norms and regulations in China. Informed consent was designed based on the NMPA regulations and ICH guidelines. Each patient who agrees to participate in the study will be invited to sign an informed consent form. Before signing the informed consent form, a specially trained researcher (understanding the key points of the study and the related risks of the study, the confidentiality of the management of records and the source of funds, etc.) will introduce the research purpose, study plan, study cycle, potential risks and expected benefits, cost and compensation, voluntary participation principle, treatment principle of study related risks, etc.

10. Randomisation

The BASIS trial will use an interactive web response system (IWRS) for central randomization stratified by study centers. Patients who sign the informed consent and meet all inclusion criteria and do not meet any exclusion criteria will be enrolled in the study. After inputting the eligible patient's necessary information into the web-based system, the researcher in each center will obtain a random code as well as the corresponding group allocation information from the IWRS. Eligible patients will be randomized into either the experimental group or the control group at a 1:1 ratio (**Figure 1**). Finally, researchers and research assistants need to print randomization application forms and center randomization lists.

The researcher team should archive the randomized list and related files (randomized application form) together in the central researcher folder (archived by the clinical study department of the center).

11. DSA and lesion evaluation

In order to ensure the quality of DSA images in the study, we will choose a clinical research center with DSA facilities. Before the start of the study, researchers from all centers will participate in the training of image generation, collection and evaluation, and obtain corresponding qualifications.

When the patient is enrolled, the central laboratory will obtain the CD disc of the patient's MRI / HRMRI imaging, CT perfusion imaging, and DSA before balloon angioplasty for central review. DSA imaging needs to show the residual vessel diameter at the maximum stenosis and the maximum length of the stenosis. If the patient underwent balloon angioplasty, arterial imaging after angioplasty should also be sent to the imaging core laboratory for central review. The standard of technical success is the residual stenosis of the target vessel $\leq 50\%$.

12. Balloon Angioplasty

Antithrombotic protocol

Pre-procedure: All patients who are scheduled to undergo balloon angioplasty should receive dual antiplatelet therapy (aspirin 100 mg per day and clopidogrel 75 mg per day) for at least three days before the procedure.

Intra-procedure: Intravenous anticoagulant such as heparin during the procedure.

Post-procedure: Aspirin 100 mg per day must be used for the entire duration of follow-up, and clopidogrel 75 mg per day or ticagrelor 90mg twice a day must be used for 90 days after randomization. If clopidogrel resistance testing is obtained and the results shows loss-of-function of the CYP2C19 allele or platelet aggregation rate of ADP $> 40\%$, then ticagrelor or cilostazole may be used as replacement. After 90 days, whether to continue to take clopidogrel will be decided according to the patient's condition and the researcher's opinion. Atorvastatin will be taken 20mg-80mg/day for one year after enrollment after which its continuation will be according the patient's condition.

396 **Anesthesia strategy and arterial access**

397 The procedure should be performed under general anesthesia. Femoral artery access is
398 recommended (radial artery access is allowed for patients with tortuosity of the aortic arch or aorta).

399 **Procedure steps**

- 400 1. A stable vascular access with a long sheath or guide catheter (an intermediate catheter is
401 recommended for participants with tortuous access), which should be placed as distal as
402 possible, is suggested to provide adequate support for performing the balloon angioplasty.
403 Collateral status assessment based on a whole brain DSA is recommended before balloon
404 angioplasty according to the ASITN/SIR collateral flow grading system (poor collateral:
405 ASITN/SIR < 3).¹²
- 406 2. An 0.014-inch microwire with a microcatheter (or not) passes through the lesion to reach the
407 distal branch of the target artery.
- 408 3. A balloon (recommending the Neuro RX and Neuro LPS Intracranial Balloon Dilation Catheter
409 [Sinomed Inc., Tianjin, China]) is navigated over the microwire to the target artery lesion.
410 Balloon size selection: the balloon length should cover the lesion completely and cover at least
411 2 mm beyond each end of the lesion. The balloon diameter is determined according to the
412 treating physician's opinion (submaximal angioplasty is suggested: balloon with a diameter no
413 more than 70% of the proximal artery diameter).
- 414 4. The balloon should be slowly inflated to nominal pressure, and maintained at this pressure for
415 10 to 30 seconds before deflating slowly. This step can be repeated 2 to 3 times if the dilation
416 effect of the balloon angioplasty is not satisfactory. For patients with extremely severe stenosis,
417 a relatively smaller balloon can be used for pre-dilation.

5. Technical success: residual stenosis $\leq 50\%$ of the proximal artery diameter with modified Thrombolysis in Cerebral Infarction (mTICI)¹³ 3 grade without arterial dissection that impairs the distal blood flow.
6. Rescue stent implantation is allowed when: (1) residual stenosis $\geq 70\%$ after balloon dilation or the antegrade flow is unstable (mTICI $< 2b$); (2) arterial dissection impairs distal blood flow (mTICI $< 2b$); (3) development of thrombosis or embolization that jeopardizes distal perfusion. Other measures such as intravenous/intra-arterial thrombolysis, infusion of GP IIb/IIIa inhibitors, are based on the physician's experience, which can be used alone or in combination with rescue stenting.
7. A final DSA should be performed after 10 to 15 minutes of observation.

13. Aggressive medical management

13.1 Antiplatelet treatment

AMM will be identical in both arms. All the necessary blood and imaging tests are collected in the screening period before randomization for the management of risk factors. Both arms are prescribed with an AMM regime (includes aspirin 100 mg per day for the entire follow-up and clopidogrel 75mg per day for the first 90 days after enrollment). Clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance (Platelet aggregation rate of adenosine diphosphate $>40\%$ or Loss-of-function allele of CYP2C19 is detected).

13.2 Risk factor management

Neurologists and study investigators at each site will be responsible for the management of the patients' risk factors. They will receive training on risk factor management according to the Chinese Stroke Association guidelines for the clinical management of cerebrovascular disorders.¹⁴ All study investigators will be regularly trained according to the BASIS protocol every 6 months. If a patient

fails to reach the target goal of risk factor management, a face-to-face follow-up with the local neurologist to modify their medical regime will be necessary.

13.2.1 Primary risk factors

Achieving target LDL cholesterol

The baseline serum low-density lipoprotein cholesterol (LDL-c) level is recorded before enrollment. The target LDL-c level is < 1.8mmol/L or 70 mg/dl. If the patient's LDL level is higher than the target level, he/she should modify the lipid-lowering drugs. Liver enzyme (AST / ALT) levels will be measured at the beginning of the study and at each visit point.

Blood pressure management

Blood pressure will be checked at screening, randomization, discharge, 30 days, 90 days, 180 days, 1 year, and at the end of the trial. Patients may need to visit local sites for blood pressure measurement and modify their medical regime if necessary. The target blood pressure is no more than 140mmHg/90mmHg for patients without DM and no more than 130mmHg/80mmHg for patients with DM. Antihypertensive drugs include Angiotensin-converting enzyme inhibitor (ACEI), Angiotensin Receptor Blocker (ARB), beta-blocker, calcium channel antagonist (CCB), diuretic, vasodilator, etc.

13.2.2 Secondary risk factors

Non-HDL cholesterol

The target level of non-HDL cholesterol is < 100 mg/dL. Non-HDL cholesterol includes low-density lipoproteins (LDL), very-low-density lipoproteins (VLDL), and intermediate-density lipoproteins (IDL). When a larger statin dosage fails to lower the LDL level, if non-HDL cholesterol $\geq$ 100 mg/dL or triglycerides are $\geq$ 200mg/dL, other lipid-lowering drugs in addition to statin are recommended, such as ezetimibe or PCSK9 inhibitors.

Diabetes management

The target of DM management is to achieve HbA1c < 7.0%. Fasting plasma glucose (FPG) and HbA1c levels will be measured at baseline, 30 days, 90 days, and 1 year.

Lifestyle modification

Smoking cessation: investigators at each site will evaluate smoking status at each follow-up and encourage all subjects to quit smoking as soon as possible.

Weight management: investigators at each site will evaluate the patient's weight according to body mass index (BMI).

Activity level: activity level for each patient is also assessed at each follow-up by trained investigators or coordinators. Moderate exercise of 30 minutes a day, three times per week is strongly recommended to all patients with athletic ability.

13.2.3 Research drugs

During the screening period of BASIS, clopidogrel will be provided free of charge for 3 to 5 days. After 90 days of enrollment, clopidogrel will also be provided free of charge. From enrollment to 1 year, atorvastatin if required by patients will be provided free of charge, and other drugs will be purchased by the patients.

At each visit point, the patient's compliance with treatment will be evaluated by counting the number of clopidogrel and atorvastatin tablets.

Patients with LDL > 1.8 mmol / L or > 70 mg / dL will be treated with atorvastatin. Whether to continue taking atorvastatin after 1 year will be assessed based on the patient's condition. If patients are taking other lipid-lowering drugs, patients are encouraged to switch to atorvastatin (not mandatory). For patients whose blood pressure does not reach the treatment goal, antihypertensive drugs will be given. Patients can choose to use but are not limited to the following antihypertensive drugs:

✓ Diuretic (Chlorthione 25mg)

- 489 ✓ Angiotensin-converting enzyme inhibitor (ACEI) (Lisinopril 10mg or 40 mg)
- 490 ✓ Angiotensin Receptor Blocker (ARB) (Candesartan 16mg or 32 mg),
- 491 ✓ Beta-blocker (Atenolol 50mg or 100mg)
- 492 ✓ Calcium channel antagonist (CCB) (Felodipine 5mg or 10 mg)
- 493 ✓ Central α agonist (clonidine 0.1mg)
- 494 ✓ Vasodilator (hydralazine 50mg)

495

496 **14. Patient and Researcher Benefit**

497 **Patient benefit:** Patients participating in this study will receive systematic treatment and follow-up.
498 Clopidogrel is provided free of charge during the screening period and 90 days of enrollment, and
499 atorvastatin tablets are provided free of charge from enrollment to 1 year. DSA is free at 1-year
500 follow-up. However, other related examinations and medications are not free, and the balloons used
501 when they are randomly assigned to the experimental group are not free.

502 **Researcher benefit:** Researchers participating in this study have the right to the use of scientific
503 data to publish scientific research.

504

505 **15. Follow-up**

506 All patients will be followed up to the primary endpoint of the study or to the end of the study.
507 Unless the patient does not agree to continue participating in the study, efforts will be made to
508 obtain complete endpoint information (whether the patient is cured or dead). If the patient
509 discontinues from this study and receives other treatments such as stenting, but agreed to provide
510 endpoint information, follow-up will continue.

All participants will be followed up by the on-site neurologists at baseline, the day of DSA, discharge, 30±7 days, 90±7 days, 6 months±14 days, 1 year ±30 days, and up to 3 years (at 6 months intervals after 1 year). Follow-up visits will be conducted by telephone at 6-month and post-1-year periods and will be evaluated in person at other visits. At each follow-up visit, the participants' medications, laboratory tests, risk factor management (as described above), and possible adverse events/endpoints are reviewed by an experienced neurologist and/or neurointerventionalist. All patients are required to undergo a neurovascular imaging examination including DSA, CTA or MRA at 1-year follow-up.

Visit 1: Screening

- ✓ Signed informed consent
- ✓ Review inclusion/exclusion criteria
- ✓ Randomization-experimental group: balloon angioplasty+AMM; control group: AMM
- ✓ Medical history, concomitant medication, demographic data collection
- ✓ Ischemic stroke history record
- ✓ Vital signs, blood routine, urine routine, blood biochemistry, coagulation function examination
- Vital signs: Blood pressure, heart rate, respiration, body temperature;
- blood biochemistry: Renal function: creatinine, urea (or urea nitrogen), glucose, uric acid, total bilirubin, direct bilirubin, indirect bilirubin, lactate dehydrogenase, creatine kinase isoenzyme;
- liver function: alkaline phosphatase, glutamyl transpeptidase, aspartate aminotransferase, alanine aminotransferase; blood lipids: total cholesterol, low-density lipoprotein, high-density lipoprotein, triglyceride;
- Coagulation function: PT, APTT, TT, FIB, INR
- ✓ Fasting blood glucose, HbA1c (optional)
- ✓ Pregnancy test

- 535 ✓ Electrocardiogram
- 536 ✓ DSA
- 537 ✓ MRI/HRMRI (HRMRI is unnecessary; if MRI is not available, CT can replace MRI)
- 538 ✓ CTP (anterior circulation sICAS)
- 539 ✓ Patient smoking cessation information, weight, and exercise will be collected. Smoking
- 540 cessation documentation will include whether the patient plans to smoke and the number of
- 541 daily cigarettes; weight: the patient's weight and BMI will reflect the patient's weight control;
- 542 exercise: weekly exercise frequency will be recorded, as well as each exercise duration
- 543 ✓ mRS, NIHSS, MoCA, EQ-5D (MoCA is unnecessary)
- 544 ✓ Platelet function test and/or CYP2C19 genotype test will be obtained according to the
- 545 availability at the center, this item is unnecessary
- 546 ✓ TCD (recommend but not necessary)
- 547 ✓ Research drug distribution: Clopidogrel (only for patients who are enrolled in BASIS)
- 548 ✓ Aspirin and clopidogrel medication orders
- 549 ✓ Concomitant medication collection
- 550 ✓ AE, SAE and endpoint observation
- 551 **Visit 2: Treatment**
- 552 ✓ Treatment according to the randomization assignment
- 553 ✓ Concomitant medication collection
- 554 ✓ Vital signs
- 555 ✓ Research drug distribution and recovery (Randomly successful patients are provided 30 days
- 556 follow-up research drug; Randomly failed patients recover research drugs issued during the
- 557 screening period; randomly successful patients are given clopidogrel and atorvastatin tablets)
- 558 ✓ Aspirin, clopidogrel and atorvastatin medication orders

- 559 ✓ Patient compliance survey
- 560 ✓ AE, SAE and endpoint observation
- 561 **Visit 3: Discharge**
- 562 ✓ Concomitant medication collection
- 563 ✓ Vital signs
- 564 ✓ Blood routine, urine routine, blood biochemistry(optional)
- 565 ✓ NIHSS
- 566 ✓ Patients' smoking cessation, weight, and exercise will be collected
- 567 ✓ Patient compliance survey
- 568 ✓ The disease treatment-related cost information collection
- 569 ✓ AE, SAE and endpoint observation
- 570 **Visit 4: 30±7 days after enrollment**
- 571 ✓ Concomitant medication collection
- 572 ✓ Vital signs, blood routine, urine routine, blood biochemistry
- 573 ✓ Fasting blood glucose, HbA1c (optional)
- 574 ✓ Patient smoking cessation, weight, and exercise data are collected
- 575 ✓ mRS
- 576 ✓ Research drug distribution: clopidogrel and atorvastatin
- 577 ✓ Aspirin, clopidogrel and atorvastatin calcium medication orders
- 578 ✓ Patient compliance survey
- 579 ✓ The disease treatment-related cost information collection
- 580 ✓ AE, SAE and endpoint observation
- 581 **Visit 5: 90±7 days after enrollment**
- 582 ✓ Concomitant medication collection

- 583 ✓ Vital signs, blood routine, urine routine, blood biochemistry
- 584 ✓ Fasting blood glucose, HbA1c (optional)
- 585 ✓ CTP (optional)
- 586 ✓ Patient smoking cessation, weight, and exercise data are collected
- 587 ✓ mRS and NIHSS
- 588 ✓ Research drug distribution: atorvastatin
- 589 ✓ Aspirin and atorvastatin medication orders
- 590 ✓ Patient compliance survey
- 591 ✓ The disease treatment-related cost information collection
- 592 ✓ AE, SAE and endpoint observation
- 593 **Visit 6: 6 months±14 days after enrollment (interview by telephone)**
- 594 ✓ Concomitant medication collection
- 595 ✓ Patient smoking cessation, weight, and exercise data are collected
- 596 ✓ mRS
- 597 ✓ AE, SAE and endpoint observation
- 598 **Visit 7: 1 year±30 days after enrollment**
- 599 ✓ Concomitant medication collection
- 600 ✓ Vital signs, blood routine, urine routine, blood biochemistry
- 601 ✓ Fasting blood glucose, HbA1c(optional)
- 602 ✓ MRI/HRMRI (HRMRI is unnecessary; if MRI is not available, CT can replace MRI)
- 603 ✓ CTP, DSA (DSA is necessary and CTP is unnecessary)
- 604 ✓ Patient smoking cessation, weight, and exercise data are collected
- 605 ✓ mRS, NIHSS, MoCA, EQ-5D (MoCA is unnecessary)
- 606 ✓ TCD (recommend but not necessary)

607 ✓ Research drug recovery: atorvastatin tablets

608 ✓ Patient compliance survey

609 ✓ Follow-up aspirin medication orders

610 ✓ The disease treatment-related cost information collection

611 ✓ AE, SAE and endpoint observation

612 **Follow-up after 1 year: every 6 months $\pm$ 30 days after 1-year enrollment**

613 Follow-up every 6 months (visit 8, visit 9, visit 10, visit 11, visit 9, and visit 11 require patients to
614 undergo face-to-face follow-up).

615 Visit by telephone:

616 ✓ Patient smoking cessation, weight, and exercise data are collected

617 ✓ Follow-up aspirin medication

618 ✓ AE, SAE and endpoint observation

619 Face-to-face visit:

620 ✓ Patient smoking cessation, weight, and exercise data are collected

621 ✓ mRS, MoCA, EQ-5D (MoCA is unnecessary)

622 ✓ Follow-up aspirin and atorvastatin tablet medication orders

623 ✓ AE, SAE and endpoint observation.

624

625 **16. Management of (transient) ischemic events during follow-up**

626 If any patient in each group has a potential clinical neurological event (transient cerebral ischemia,
627 ischemic stroke), the neurologist will determine whether to perform a vessel imaging examination,
628 including DSA. The most effective treatment for ICAS patients with transient cerebral ischemia or
629 ischemic stroke is unknown. The researchers will determine the best treatment according to the
630 patient's condition and actively control the risk factors.

If patients in the experimental group experience transient cerebral ischemia or ischemic stroke, accompanied by restenosis $\geq 70\%$ diagnosed by DSA, investigators composed of neurologists and neurointerventional experts will provide advice on whether the patient should undergo revascularization or medical management alone. Revascularization is not recommended for patients with restenosis $< 70\%$.

17. Assessment of endpoint and adverse events (AEs)

17.1 Endpoint event assessment

The endpoint events include target vessel haemorrhagic stroke, ischemic stroke, death, and target vessel revascularization. If a patient has a potential primary endpoint event, the researcher must report to a third-party monitoring company project team within 48 hours, including the patient's complete medical records. The project team will submit the patient's data to the event arbitration committee, and the event arbitration committee will complete the arbitration within 7 days. The endpoint event arbitration committee is composed of three members. First, two members use back-to-back arbitration. If the evaluation opinions are inconsistent, a third evaluation expert will cast a decisive vote.

17.2 AEs assessment

An AE is the presence of any unexpected medical condition during or after being treated with medical devices. It includes symptoms, signs, or abnormal laboratory parameters that can be unrelated to treatment. An AE that fulfills at least one of the following criteria is classified as a severe AE(SAE): AE that results in death; is life-threatening; requires/prolongs inpatient hospitalization; results in serious disability; or requires medical intervention to prevent one of the above-mentioned outcomes. If a potential endpoint occurs, the committee board meeting will be convened to evaluate whether such an event can be categorized as the primary endpoint.

Unanticipated adverse device effect (UADE): Any serious adverse reaction affecting health or safety, or any life-threatening event or death related to (caused by) the device. The nature, severity, or degree of such serious adverse reactions, life-threatening events or deaths, or other unexpected device-related serious events involving patient rights, safety, or welfare, were not previously identified in the study plan.

The following events are considered as possible (expected) complications of balloon angioplasty:

Vasospasm	Neurological symptoms	Coagulation disorders
Dissection	Pseudoaneurysm	Death
Vessel occlusion	Renal insufficiency/failure	Embolism (air, tissue, thrombus or foreign body)
Vessel perforation	Respiratory distress/failure hematoma	Pain, and/or puncture site infection
Vessel rupture	Seizure	Hemorrhage
Vessel embolism	Infection	Hypertension
Stroke	Drug reaction of antiplatelets	Hypotension
Transient ischemic attack	Syncope	

17.3 SAE record and report

The collection time of adverse events and serious adverse events in the study begins with signing the informed consent, across the study period, until the end of the study. SAEs of patients who discontinue treatment in advance will be recorded.

When any SAE occurs during the study, the researcher should immediately treat the patient, and report in writing within 24 hours to the clinical trial institution, ethics committee, and sponsor (Beijing Tiantan Hospital, Capital Medical University). For death events, clinical trial institutions and researchers should provide the ethics committee and sponsors (Beijing Tiantan Hospital, Capital Medical University) with all the required information.

17.4 Prediction of AEs and countermeasures

671 AEs that may occur during the procedures include, but are not limited to:

672 **Thromboembolic events**

673 The possible causes include no anticoagulation or insufficient anticoagulation, insufficient
674 antiplatelet aggregation treatment before and after the procedure or aspirin resistance.
675 Anticoagulation and antiplatelet aggregation should be reasonably standardized. If embolization
676 occurs in the important vessels, it should be handled by a professional doctor and assessed
677 according to the patient's condition.

678 **Cerebral vasospasm**

679 Cerebral vasospasm can be caused by subarachnoid hemorrhage (SAH), or by instrumentation with
680 endovascular catheters and guidewires. If vasospasm occurs, the professional doctor will give
681 symptomatic treatment per the patient's condition.

682 **Acute vessel occlusion**

683 Possible causes include intimal injury, platelet aggregation, thrombosis, etc., usually occurring
684 within 30 minutes after balloon angioplasty or may be related to intimal avulsion. Preventive
685 measures: including aspirin, ticlid or clopidogrel antiplatelet drugs can effectively reduce acute
686 thrombotic events. Under strict pre-procedure antiplatelet therapy and intraprocedure
687 anticoagulation, acute vessel occlusion events can be avoided.

688 Treatment measures: Once acute vessel occlusion occurs, reasonable treatment should be performed
689 immediately. Thrombolysis can be performed in patients with acute thrombosis.

690 **Thrombosis**

691 Thrombosis occurs mainly due to plaque fragmentation caused by squeezing the plaque during
692 balloon dilatation. Sometimes, thrombosis is caused by the catheter and guide wire. The use of brain
693 protection devices and intraprocedure anticoagulation are the basic measures to prevent this event.

694 **Vessel rupture**

A vessel rupture complication can directly lead to death. When balloon angioplasty is performed, the selected balloon is preferably slightly smaller than the diameter of the target vessel. Once vessel rupture occurs, symptomatic treatment should be performed immediately by a professional physician.

Cerebral hyperperfusion syndrome

For severe stenotic lesions, the autoregulation ability of distal vessels decreases due to long-term ischemia. The sudden expansion of the lesion site diameter may significantly increase the blood flow. The distal vessels, especially the cortical arteries, cannot immediately adapt to this sudden increase in blood flow and passively expand, resulting in cerebral hyperperfusion syndrome. The clinical presentation includes headache, nausea, vomiting, seizure, and disturbance of consciousness. Some patients can suffer intracranial hemorrhage. The possibility of this occurrence should be fully recognized before the procedure, and blood pressure should be appropriately reduced during and after the procedure to prevent this event. If postoperative cerebral hyperperfusion syndrome occurs, symptomatic treatment should be performed according to the patient's condition (such as antihypertensive medications, analgesics, antiepileptic drugs), agents to reduce intracranial pressure, surgical treatment of cerebral hemorrhage, etc.

18. Core Laboratory

Imaging will be adjudicated by an independent imaging core lab (China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence). All of the blood samples and imaging data will be collected, transported, and preserved according to the study protocol. Imaging will be blinded (clinical features and treatment options for patients), reviewed by experienced MRI experts and DSA experts (over a 10-year working experience).

19. Quality control

To ensure that the collected data is concordant with the protocol, each sub-center of the BASIS trial will be regularly scrutinized, and by comparing data in the Electronic Data Capture System (EDC) and data in the original documents (case report form [CRF] vs. source documents) to confirm data consistency. If a patient develops an AE/severe AE or completed a 1-year observation after receiving treatment from non-study sites, duplicate copies of the medical documents will be collected for future review.

20. Statistical Analysis

The composite event rates and corresponding 95% CIs of the primary outcome in the two treatment groups will be estimated by Kaplan-Meier survival analysis and compared by log-rank test. A hazard ratio between the two groups and its 95% CI will be calculated using a Cox proportional hazards regression model. Time-to-event endpoints of the secondary outcomes will also be analyzed using Kaplan-Meier survival analysis and Cox regression, and common odds ratios of mRS will be estimated using ordinal logistic regression. The widths of the intervals will not be adjusted for multiplicity for secondary outcomes. The main analysis of this study will be conducted based on the intention-to-treat principle, and per protocol analysis and as treated analysis will also be conducted as sensitivity analyses. All analyses will be performed using SAS software, Version 9.4 (SAS Institute). Detailed statistical methods, missing data imputation and subgroup analyses are described in the statistical analysis plan.

20.1 Sample size

The estimation of the sample size is calculated according to the 1-year results of the SAMMPRIS trial, the incidence of target vessel stroke or death rate was 12.2% within 12 months after

enrollment in the experimental group.¹ The sample size was 686 cases, with a 80% power, two-sided $\alpha = 0.05$, to detect a 50 % relative reduction (6.1%) after balloon angioplasty for intracranial atherosclerosis. It is assumed that the O'Brien-Fleming spending function will lead to 6% expansion of the total type I error. The SAS software is used to calculate the sample size required for the mid-term analysis, which increases the sample size to the required 728 cases. Considering the 10% lost to follow-up rate, the final sample size of this study is 802 cases.

20.2 Interim analysis

Interim analysis is performed when half of patients finished 12-months visit. The mid-term analysis α_1 is estimated to be 0.0015, and the final analysis α_2 is 0.024.

20.3 Subgroup analysis

Subgroup analyses on the primary outcome will be performed in the following subgroups:

- ✓ Age, <65 years old vs. ≥ 65 years old
- ✓ Sex, men vs. women
- ✓ Hypertension, yes vs. no
- ✓ Diabetes, yes vs. no
- ✓ Smoking, yes vs. no
- ✓ Baseline renal function, eGFR <60 ml/min/1.73m² vs. eGFR ≥ 60 ml/min/1.73m²
- ✓ Target vessel stenosis degree, <80% vs. $\geq 80\%$
- ✓ BMI, <25 Kg/m² vs. 25-30 Kg/m² vs. ≥ 30 Kg/m²
- ✓ Hypoperfusion, yes vs. no
- ✓ Lesion location, anterior circulation vs. posterior circulation
- ✓ The mechanism, ischemic stroke vs. TIA

21. Data Safety and Monitoring Board

The data safety and monitoring board (DSMB) is comprised of academic members and an independent statistician. The DSMB is scheduled to have meetings annually to review the study progress to guarantee the trial is in accordance with the standards of ethics and ensure the safety of all enrolled patients. After each DSMB meeting, the DSMB members will generate a report including all recommendations, and hand it to the steering committee immediately after the meeting.

22. Study organization

Twice a year, the steering committee will meet to oversee the trial and provide strategic guidance. Periodically, the clinical research team of the leading center will meet online with the project team of the clinical research organization to review the trial progress and data monitoring each week. An independent clinical events adjudication committee (CEC) will ensure that defined outcomes are reported and judged uniformly using the same definition by experts who are blinded to the treatment status. (Table 1)

Table 1. Collaboration team of trial and Study Committee

Collaboration team	Unit or Hospital name	Investigator Name
Sponsor	Beijing Tian Tan Hospital, Capital Medical University	NA
CRO	Beijing Jingchengtong Medicine Technology Co., Ltd.	NA
Statistical analysis	Department of Biostatistics, Peking University Clinical Research Institute	NA
Imaging core lab	China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence	Jing Jing
	China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence	Nan Qi

Collaboration team	Unit or Hospital name	Investigator Name
CEC team	Tangdu Hospital of Air Force Medical University	Zhenwei Zhao
	Beijing Tian Tan Hospital, Capital Medical University	Kehui Dong
	The First Affiliated Hospital of Harbin Medical University	Huaizhang Shi
DSMB	People's Liberation Army Air Force General Hospital	Jin Shi
	Strategic Support Force Specialty Medical Center	Yiling Cai
	China National Clinical Research Center for Neurological Diseases	Yuesong Pan
AE Committee	Beijing Tian Tan Hospital, Capital Medical University	Xuan Sun
	Beijing Tian Tan Hospital, Capital Medical University	Yiming Deng
	Beijing Tian Tan Hospital, Capital Medical University	Ning Ma
Steering Committee	Beijing Tian Tan Hospital, Capital Medical University	Zhongrong Miao
	Beijing Tian Tan Hospital, Capital Medical University	Yilong Wang
	Beijing Tian Tan Hospital, Capital Medical University	Yongjun Wang
	Comprehensive Stroke Center, University of California, Irvine, USA	Wengui Yu
	Beijing Tian Tan Hospital, Capital Medical University	Xingquan Zhao
	Neurovascular Imaging Research Core and Department of Neurology, University of California, USA	David S Liebeskind
	Department of Neurosurgery, University Hospitals Cleveland Medical Center/ Case Western Reserve University School of Medicine, Ohio, USA	Sepideh Amin-Hanjani
Cooperation team	Unit or Hospital name	Investigator Name
Sponsor	Beijing Tian Tan Hospital, Capital Medical University	NA
CRO	Beijing Jingchengtong Medicine Technology Co., Ltd.	NA
Statistical analysis	Medical Statistics Office, Peking University First Hospital, Beijing, China	NA
Imaging core lab	China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence	Jing Jing Nan Qi
CEC team	Tangdu Hospital of Air Force Medical University	Zhenwei Zhao
	Beijing Tian Tan Hospital, Capital Medical University	Kehui Dong
	The First Affiliated Hospital of Harbin Medical University	Huaizhang Shi
DSMB	People's Liberation Army Air Force General Hospital	Jin Shi
	Strategic Support Force Specialty Medical Center	Yiling Cai
	Beijing Tian Tan Hospital, Capital Medical University	Yuesong Pan
AE Committee	Beijing Tian Tan Hospital, Capital Medical University	Xuan Sun
	Beijing Tian Tan Hospital, Capital Medical University	Yiming Deng
	Beijing Tian Tan Hospital, Capital Medical University	Ning Ma
Technology Steering Committee	Strategic Support Force Specialty Medical Center	Yiling Cai
	Xuanwu Hospital Capital Medical University	Yabing Wang
	Aerospace Central Hospital	Peifu Wang
	Beijing Chaoyang Emergency Rescue Hospital of	Qiang Zhang

Collaboration team	Unit or Hospital name	Investigator Name
	Integrated Traditional Chinese and Western Medicine The Sixth Medical Center of the General Hospital of the People's Liberation Army	Feng Qiu
Chairs of	Beijing Tian Tan Hospital, Capital Medical University	Zhongrong Miao
Steering	Beijing Tian Tan Hospital, Capital Medical University	Yilong Wang
Committee	Comprehensive Stroke Center, University of California, Irvine, USA	Wengui Yu

23. Research drug supply and responsibilities division

Clopidogrel will be provided to the patients for free during the screening period of 3-5 days and the enrollment period of 90 days. Whether to continue taking clopidogrel after 90 days will be assessed according to the patient's condition. The patient will need to purchase clopidogrel after 90 days.

Atorvastatin tablets are also free for 1 year. Whether to continue taking atorvastatin tablets after 1 year will be assessed according to the patient's condition. Patients will need to buy atorvastatin after 1 year.

Drugs used in BASIS will be stored in a safe place by researchers or other authorized personnel and stored in accordance with the conditions required (the drug instructions).

Researchers, hospital pharmacists or other persons are permitted to store and distribute the research drugs, and are responsible for ensuring that the drugs used in clinical trials are safely maintained in accordance with the requirements of the (research) organizers and meet the corresponding regulatory requirements.

All drugs used in clinical studies should be distributed according to the researcher's prescription, and the researcher should ensure that the drugs used for research distribution and recovery are accurately recorded.

Any quality problems related to research drugs (labels, failure dates, etc.) encountered in the reception or use should be reported to the organizer in time, and the organizer should initiate a complaint procedure.

Under no circumstances will researchers provide research results to third parties, allow the research results to be used under non-clinical research protocols, or transfer the research results in any other way.

24. Research Device Processing Mechanism, Characteristics, and Responsibilities Division

Balloon devices provided by SINOMED will be used in accordance with high-quality clinical practice guidelines and medical device tracking guidelines. An introduction to the balloon device contains the following information:

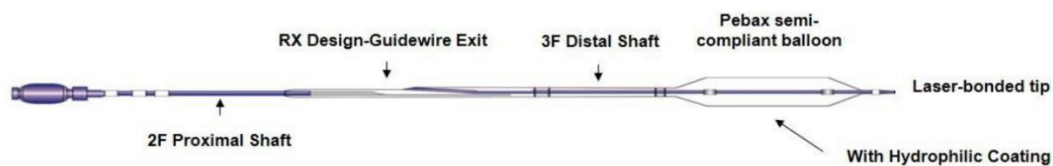
Device name: Neuro RX intracranial balloon catheter

Research and development unit: Sino Medical Science and Technology Co., Ltd.

Registration number: CFDA, 20163772491 (China, Equipment, Registration, Approval, 2016-12-19)

Composition: Neuro RX intracranial balloon catheter is an expandable balloon with a rapid exchange system that can be used to treat ICAS. Its working length is 145 cm, with a guide wire of less than 0.014' and a catheter with an inner diameter of $\geq 5F$ (0.056'). Two opaque markers on the catheter shaft indicate the working length of the balloon. The guide wire exit is 24 cm from the far tip.

Device parameters (length and diameter)



Length Diameter	6 mm	10 mm	12 mm	15 mm	20 mm	25 mm
1.50		✓		✓	✓	
2.00		✓		✓	✓	✓
2.25		✓		✓	✓	✓
2.50	✓	✓	✓	✓	✓	✓
2.75	✓	✓	✓	✓	✓	✓
3.00	✓	✓	✓	✓	✓	✓
3.25	✓	✓	✓	✓	✓	✓
3.50	✓	✓	✓	✓	✓	✓
4.00	✓	✓	✓	✓	✓	✓

Recommend: The Neuro RX intracranial balloon catheter is suitable for ICAS lesion, and balloon dilatation for improving intracranial perfusion.

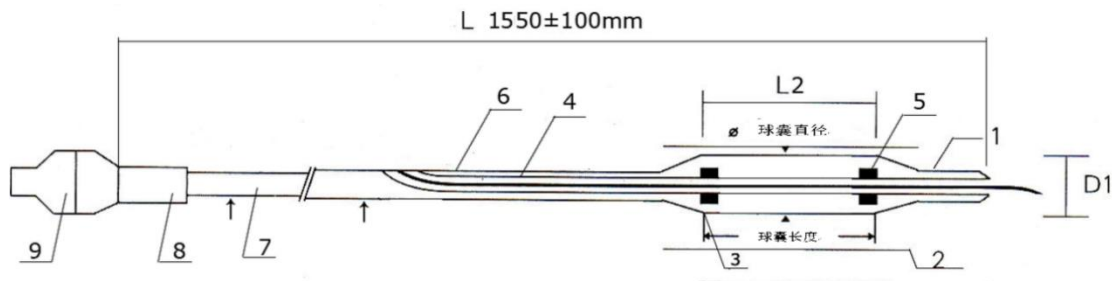
Device name: Neuro LPS intracranial balloon dilatation catheter

Research and development unit: Sino Medical Science and Technology Co., Ltd.

Registration number: NMPA, 20203030576 (China, Equipment, Registration, Approval, 2020-06-19)

Composition:

Neuro LPS intracranial balloon dilatation catheter is composed of medical polypropylene, polytetrafluoroethylene, polyether amide block copolymer, polycarbonate, polyolefin, stainless steel, and platinum alloy.



835

836 1) TIP head (polyether amide block copolymer)

837 2) Balloon protective sheath (polytetrafluoroethylene)

838 3) Balloon (polyetheramide block copolymer)

839 4) Balloon inner tube (polyetheramide block copolymer)

840 5) Marker (two Markers are respectively set on the balloon's inner tube (4), the outer edge of the
 841 two Markers is flush with the neck of the balloon, and the material is 90% platinum and 10%
 842 iridium alloy)

843 6) Balloon outer tube (polyetheramide block copolymer)

844 7) Metal conduit (304 stainless steel)

845 8) Sheath (two layers, inner layer of 304 stainless steel, outer layer of polyolefin)

846 9) Joint (polycarbonate)

847 Note 1: D1 represents the diameter of the balloon after expansion.

848 Note 2: L2 represents the effective length of the balloon on the balloon catheter.

849 Note 3: L represents the effective length of the balloon catheter.

850 Note 4: A complete balloon dilatation catheter is inserted into a high-density polyethylene ring protective sleeve.

851 Device parameters (length and diameter)

Length Diameter	8 mm	10 mm	12 mm	15 mm	20 mm
1.50	●	●	●	●	●
2.00	●	●	●	●	●
2.25	●	●	●	●	●
2.50	●	●	●	●	●
2.75	●	●	●	●	●
3.00	●	●	●	●	●
3.25	●	●	●	●	●
3.50	●	●	●	●	●
4.00	●	●	●	●	●

Description: The Neuro RX or Neuro LPS intracranial balloon dilatation catheter is provided by SINOMED. Intracranial balloon catheter equipment is not provided free of charge.

Responsibilities Division

Investigators and/or other designated persons/institutions should archive and save the delivery records of the products to the research center, including the inventory of the research center, the use of each item, and the unused products that will be returned to the sponsor. In the event of damage, loss, or additional (unscheduled) material, the organizer or relevant party shall be notified. These records should include the date, quantity, product serial number, shelf life (if any), and a unique code number assigned to the device and the research object.

25. Control requirements

BASIS will be carried out in accordance with the relevant provisions of the China Food and Drug Administration (NMPA): 'medical device clinical trial quality management norms' and 'drug clinical trial quality management norms'.

26. Institutional Review Board (IRB)/Ethics Committee (IEC)

Researchers or organizers must submit a clinical research protocol to the Ethics Committee of Beijing Tiantan Hospital and obtain a written certificate signed by the Ethics Committee with clear approval time and approval opinions.

The following information should be clearly marked on the written certificate of approval: clinical trial information (study number, name, and version number of clinical study protocol), review documents (study protocol, informed consent, etc.).

The clinical research will be initiated only after the research sponsor has received written evidence of a clear approval time and approval opinion.

During the clinical research, any revision or modification of the clinical study protocol, any event that may affect patient safety or the continuation of clinical research, especially any safety changes, and any updates to the Investigator's Manual should be reported to the Ethics Committee.

If necessary, a progress report is provided to the ethics committee each year, and the clinical research results are summarized at the end of the clinical study.

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Appendix

Appendix Table 1. ASITN/SIR collateral circulation scale

Grade 0	No collaterals visible to the ischemic site
Grade 1	Slow collaterals to the periphery of the ischemic site with persistence of some of the defect
Grade 2	Rapid collaterals to the periphery of ischemic site with persistence of some of the defect and to only a portion of the ischemic territory
Grade 3	Collaterals with slow but complete angiographic blood flow of the ischemic bed by the late venous phase
Grade 4	Complete and rapid collateral blood flow to the vascular bed in the entire ischemic territory by retrograde perfusion

Appendix Table 2. modified Rankin Scale

The modified Rankin Scale (mRS) is an ordinal hierarchical scale ranging from 0 to 5, with higher scores indicating more severe disability. A score of 6 has been added to signify death.

Category	Short description	Long description
0	No symptoms	No symptoms
1	Symptoms, no disability	Minor symptoms that do not interfere with lifestyle
2	Slight disability	Slight disability, symptoms that lead to some restriction in lifestyle, but do not interfere with the patient's capacity to look after himself.
3	Moderate disability	Moderate disability, symptoms that significantly restrict lifestyle and prevent totally independent existence
4	Moderately severe disability	Moderately severe disability, symptoms that clearly prevent independent existence though not needing constant attention
5	Severe disability	Severe disability, totally dependent patient requiring constant attention day and night.
6	Death	Death

975 **Appendix Table 3. National Institute of Health Stroke Scale (NIHSS)**

976 The NIHSS is an ordinal hierarchical scale to evaluate the severity of stroke by assessing a
 977 patient's performance. Scores range from 0 to 42, with higher scores indicating a more severe
 978 deficit. Administer stroke scale items in the order listed. Record performance in each category after
 979 each subscale exam. Do not go back and change scores. Follow directions provided for each exam
 980 technique. Scores should reflect what the patient does, not what the clinician thinks the patient can
 981 do. The clinician should record answers while administering the exam and work quickly. Except
 982 where indicated, the patient should not be coached (i.e. repeated requests to patient to make a
 983 special effort).

Instructions	Scale definition
1a. Level of consciousness. The investigator must choose a response if a full evaluation is prevented by such obstacles as an endotracheal tube, language barrier, orotracheal trauma/bandages. A 3 is scored only if the patient makes no movement (other than reflexive posturing) in response to noxious stimulation.	0 = Alert; keenly responsive. 1 = Not alert; but arousable by minor stimulation to obey, answer, or respond. 2 = Not alert; required repeated stimulation to attend, or is obtunded and requires strong or painful stimulation to make movements (not stereotyped). 3 = Responds only with reflex motor or autonomic effects or totally unresponsive, flaccid and areflexic.
1b. LOC Questions: The patient is asked the month and his/her age. The answer must be correct – there is not partial credit for being close. Phasic and stuporous patients who do not comprehend the questions will score 2. Patients unable to speak because of endotracheal intubation, orotracheal trauma, severe dysarthria from any cause, language barrier, or any other problem not secondary to aphasia are given a 1. It is important that only the initial answer be graded and that the examiners not “help” the patient with verbal or non-verbal clues.	0 = Answers both questions correctly. 1 = Answers one question correctly. 2 = Answers neither question correctly.
1c. LOC Commands: The patient is asked to open and close the eyes and then to grip and release the non-paretic hand. Substitute another one step command if the hand cannot be used. Credit is given if an unequivocal attempt is made but not completed due to weakness. If the patient does not respond to command, the task should be demonstrated to him or her (pantomime), and the result scored (i.e. follows none, one or two commands). Patients with trauma, amputation, or other physical impediments should be given suitable one-step commands. Only the first	0 = Performs both tasks correctly. 1 = Performs one task correctly. 2 = Performs neither task correctly.

attempt is scored.	
2. Best Gaze: Only horizontal eye movements will be tested. Voluntary or reflexive (oculocephalic) eye movements will be scored, but caloric testing is not done. If the patient has a conjugate deviation of the eyes that can be overcome by voluntary or reflexive activity, the score will be a 1. If a patient has an isolated peripheral nerve paresis (CN III, IV or VI), score a 1. Gaze is testable in all aphasic patients. Patients with ocular trauma, bandages, preexisting blindness, or other disorder of visual acuity or fields should be tested with reflexive movements, and a choice made by the investigator. Establishing eye contact and then moving about the patient from side to side will occasionally clarify the presence of a partial gaze palsy.	0= Normal. 1= Partial gaze palsy; gaze is abnormal in one or both eyes, but forced deviation or total gaze paresis is not present. 2= Forced deviation; or total gaze paresis not overcome by the oculocephalic maneuver.
3. Visual: Visual fields (upper and lower quadrants) are tested by confrontation, using finger counting or visual threat, as appropriate. Patients may be encouraged, but if they look at the side of the moving finger appropriately, this can be scored as normal. If there is unilateral blindness or enucleation, visual fields in the remaining eye are scored. Score 1 only if a clear-cut asymmetry, including quadrantanopia, is found. If patient is blind from any cause, score 3. Double simultaneous stimulation is performed in this case. If there is extinction, the patient receives a 1, and the results are used to respond to item 11.	0= No visual loss. 1= Partial hemianopia. 2= Complete hemianopia. 3= Bilateral hemianopia (blind including cortical blindness)
4. Facial palsy: Ask or use pantomime to encourage the patient to show teeth or raise eyebrows and close eyes. Score symmetry of grimace in response to noxious stimuli in the poorly response or non-comprehending patient. If facial trauma/bandages, orotracheal tube, tape or other physical barriers obscure the face, these should be removed to the extent possible.	0 = Normal symmetrical movements. 1= Minor paralysis (flattened nasolabial fold, asymmetry on smiling) 2= Partial paralysis (total or near-total paralysis of lower face) 3= Complete paralysis of one or both sides (absence of facial movement in the upper and lower face).
5. Motor arm: The limb is placed in the appropriate position: extend the arms (palms down) 90 degrees (if sitting) or 45 degrees (if supine). Drift is scored if the arm falls before 10 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic arm. Only in the case of amputation or joint fusion at the shoulder, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	0= No drift; limb holds 90 (or 45) degrees for full 10 seconds. 1= Drift; limb holds 90 (or 45) degrees, but drifts down before full 10 seconds; does not hit bed or other support. 2= Some effort against gravity; limb cannot get to or maintain (if cued) 90 (or 45) degrees, drifts down to bed, but has some effort against gravity. 3= No effort against gravity; limb falls. 4= No movement.

	UN = Amputation or joint fusion: explain: 5a = Left Arm. 5b = Right arm.
6. Motor leg: The limb is placed in the appropriate position: hold the leg at 30 degrees (always tested supine). Drift is scored if the leg falls before 5 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic leg. Only in the case of amputation or joint fusion at the hip, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	0= No drift; leg holds 30-degree position for full 5 seconds. 1= Drift; leg falls by the end of the 5-second period but does not hit bed. 2= Some effort against gravity; leg falls to bed by 5 seconds, but has some effort against gravity. 3= No effort against gravity; leg falls to bed immediately. 4= No movement. UN = Amputation or joint fusion: explain: 6a. Left Leg 6b. Right Leg.
7. Limb ataxia: This item is aimed at finding evidence of a unilateral cerebellar lesion. Test with eyes open. In case of visual defect, ensure testing is done in intact visual field. The finger-nose-finger and heel-shin tests are performed on both sides, and ataxia is scored only if present out of proportion to weakness. Ataxia is absent in the patient who cannot understand or is paralyzed. Only in the case of amputation or joint fusion, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice. In case of blindness, test by having the patient touch nose from extended arm position.	0= Absent. 1= Present in one limb. 2= Present in two limbs. UN = Amputation or joint fusion: explain:
8. Sensory: Sensation or grimace to pinprick when tested, or withdrawal from noxious stimulus in the obtunded or aphasic patient. Only sensory loss attributed to stroke is scored as abnormal and the examiner should test as many body areas (arms [not hands], legs, trunk, face) as needed to accurately check for hemisensory loss. A score of 2, 'severe or total sensory loss', should only be given when a severe or total loss of sensation can be clearly demonstrated. Stuporous and aphasic patients will, therefore, probably score 1 or 0. The patient with brainstem stroke who has bilateral loss of sensation is scored 2. If the patient does not respond and is quadriplegic, score 2. Patients in a coma (item 1a=3) are automatically given a 2 on this item.	0= Normal; no sensory loss. 1= Mild-to-moderate sensory loss; patients feels pinprick is less sharp or is dull on the affected side; or there is a loss of superficial pain with pinprick, but patient is aware of being touched. 2= Severe to total sensory loss; patient is not aware of being touched in the face, arm and leg.
9. Best language: A great deal of information about comprehension will be obtained during the preceding	0= No aphasia; normal 1= Mild-to-moderate aphasia; some

984

sections of the examination. For this scale item, the patient is asked to describe what is happening in the attached picture, to name the items on the attached naming sheet and to read from the attached list of sentences. Comprehension is judged from responses here, as well as to all of the commands in the preceding general neurological exam. If visual loss interferes with the tests, ask the patient to identify objects placed in the hand, repeat, and produce speech. The intubated patient should be asked to write. The patient in a coma (item 1a=3) will automatically score 3 on this item. The examiner must choose a score for the patient with stupor or limited cooperation, but a score of 3 should be used only if the patient is mute and follows no one-step commands.	obvious loss of fluency or facility of comprehension, without significant limitation on ideas expressed or form of expression. Reduction of speech and/or comprehension, however, makes conservation about provided materials difficult or impossible. For example, in conversation about provided materials, examiner can identify picture or naming card content from patient's response. 2= Severe aphasia; all communication is through fragmentary expression; great need for inference, questioning, and guessing by the listener. Range of information that can be exchanged is limited; listener carries burden of communication. Examiner cannot identify materials provided from patient response. 3 = Mute, global aphasia: no usable speech or auditory comprehension.
10. Dysarthria: If patient is thought to be normal, an adequate sample of speech must be obtained by asking patient to read or repeat words from the attached list. If the patient has severe aphasia, the clarity of articulation of spontaneous speech can be rated. Only if patient is intubated or has other physical barriers to producing speech, the examiner should record the score as untestable (UN), and clearly write an explanation for this choice. Do not tell the patient why he or she is being tested.	0= Normal. 1= Mild-to-moderate dysarthria; patient slurs at least some words and, at worst, can be understood by some difficulty. 2= Severe dysarthria: patient's speech is so slurred as to be unintelligible in the absence of or out of proportion to any dysphasia, or is mute/anarthric. UN = Intubated or other physical barrier.
11. Extinction and Inattention (formerly Neglect): Sufficient information to identify neglect may be obtained during the prior testing. If the patient has a severe visual loss preventing visual double simultaneous stimulation, and the cutaneous stimuli are normal, the score is normal. If the patient has aphasia but does appear to attend to both sides, the score is normal. The presence of visual spatial neglect or anosagnosia may also be taken as evidence of abnormality. Since the abnormality is scored only if present, the item is never untestable.	0= No abnormality. 1= Visual, tactile, auditory, spatial, or personal inattention or extinction to bilateral simultaneous stimulation in one of the sensory modalities. 2= Profound hemi-inattention or extinction to more than one modality; does not recognize own hand or orients to only one side of space.

985 **Appendix Table 4. EuroQoL 5D-5L**

986 Under each heading, please tick the ONE box that best describes your health TODAY.

987 **Mobility**

I have no problems in walking about ☐

I have slight problems in walking about ☐

I have moderate problems in walking about ☐

I have severe problems in walking about ☐

I am unable to walk about ☐

Self-care

I have no problems washing or dressing myself ☐

I have slight problems washing or dressing myself ☐

I have moderate problems washing or dressing myself ☐

I have severe problems washing or dressing myself ☐

I am unable to wash or dress myself ☐

Usual activities (e.g. work, study, housework, family or leisure activities)

I have no problems doing my usual activities ☐

I have slight problems doing my usual activities ☐

I have moderate problems doing my usual activities ☐

I have severe problems doing my usual activities ☐

I am unable to do my usual activities ☐

Pain/discomfort

I have no pain or discomfort ☐

I have slight pain or discomfort ☐

I have moderate pain or discomfort ☐

I have severe pain or discomfort ☐

I have extreme pain or discomfort ☐

Anxiety/depression

I am not anxious or depressed ☐

I am slightly anxious or depressed ☐

I am moderately anxious or depressed ☐

I am severely anxious or depressed ☐

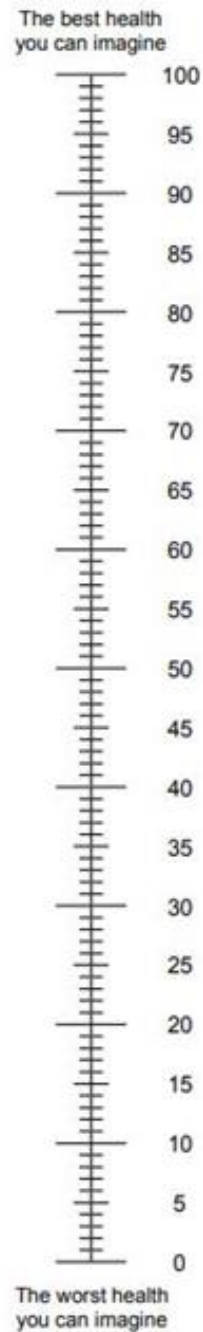
I am extremely anxious or depressed ☐

988

- We would like to know how good or bad your health is TODAY.
- This scale is numbered from 0 to 100.
- 100 means the best health you can imagine.
- 0 means the worst health you can imagine.
- Mark an X on the scale to indicate how your health is TODAY.
- Now, please write the number you marked on the scale in the box below.

YOUR HEALTH TODAY =

989



Appendix Table 5. mTICI grade

mTICI grade	Definitions
0 grade	No perfusion
1 grade	Antegrade reperfusion past the initial occlusion, but limited distal branch filling with little or slow distal reperfusion
2a grade	Antegrade reperfusion of less than half of the occluded target artery previously ischemic territory (eg, in 1 major division of the MCA and its territory)
2b grade	Antegrade reperfusion of more than half of the previously occluded target artery ischemic territory (eg, in 2 major divisions of the MCA and their territories)
3 grade	Complete antegrade reperfusion of the previously occluded target artery ischemic territory, with absence of visualized occlusion in all distal branches

**Balloon Angioplasty vs. Aggressive Medical Management for
Symptomatic Intracranial Artery Stenosis: The BASIS Randomized
Clinical Trial
(BASIS)**



Protocol

Research team: Beijing Tiantan Hospital, Capital Medical University, Beijing, China

Principal Investigator: Zhongrong Miao, MD, Professor of Neurology

Co-Principal Investigator: Yilong Wang, MD, Professor of Neurology

Protocol Version 2.3

Date: March 28, 2022

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Protocol Signature Page

I have read this protocol and agree to adhere to the requirements.

By signing this document we confirm that the clinical study will be conducted in accordance with the protocol and all applicable laws and regulations including, but not limited to, the International Conference on Harmonisation Guideline for Good Clinical Practice (GCP) and the ethical principles that have their origins in the Declaration of Helsinki.

Clinical Site

Site Principal Investigator Signature

Date

1. BASIS Protocol Synopsis

Official Title	Balloon Angioplasty for Symptomatic Intracranial Artery Stenosis: A Multicenter, Prospective, Open-Label, Blinded-Endpoint, Randomized, Controlled Trial
National Subject and Number	Prediction model and intervention strategy for recurrence and progression of symptomatic intracranial and extracranial major artery stenosis (2017YFC1307900)
Acronym	BASIS
PI	Prof. Zhongrong Miao and Prof. Yilong Wang
Study Objectives	To evaluate whether combined balloon angioplasty and aggressive medical management (AMM) is superior to AMM for patients with high-risk intracranial atherosclerotic stenosis (ICAS)
Study Design	The BASIS trial is an investigator-initiated, multicenter, prospective, randomized, open-label, blinded end-point (PROBE) trial Patients eligible in BASIS are 35-80 years old, with a recent transient ischemic attack (<90 days) or ischemic stroke (14-90 days) before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial artery (terminal internal carotid artery, middle cerebral artery M1 segment, vertebral artery V4 segment, and basilar artery) and are randomly assigned to an experimental group and a control group Experimental group: patients will receive balloon angioplasty plus AMM; Control group: patients will receive the AMM alone ; ✓ AMM : aspirin 100 mg daily for the entire follow-up and clopidogrel 75mg daily for the first 90 days after enrollment. Whether to continue taking clopidogrel after 90 days will be assessed according to the condition of the patient, clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance; atorvastatin will

	be taken 20mg-80mg/day for one year after enrollment; the dose was adjusted to control LDL <1.8mmol/L or <70mg/dl. Whether to continue taking atorvastatin after one year will be assessed according to the patient's condition. ✓ Risk factor management—Control blood pressure 130-140/90-100 mmHg
Primary Endpoint	The primary outcome is stroke or death within 30 days after enrollment, or any ischemic stroke or revascularization of the qualifying artery beyond 30 days through 12 months after enrollment.
Secondary Endpoints	- Any stroke (ischemic or hemorrhage stroke) or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up - Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 90 days after enrollment - Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 90 days after enrollment - mRS (modified Rankin Scale) at 90 days - Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 12 months after enrollment - Revascularization of the qualifying artery within 12 months after enrollment - Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 12 months after enrollment - mRS at 12 months - Restenosis rate of the target artery within 12 months (defined as rate of stenosis >70% or increased by 30% based on following-up neurovascular imaging) - Combined events such as stroke, myocardial infarction, and vascular death within 12 months after enrollment - Quality of life assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire) within 12 months after enrollment

	12. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 24 months after enrollment 13. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 24 months after enrollment 14. mRS at 24 months 15. Combined events such as stroke, myocardial infarction, and vascular death within 24 months after enrollment 16. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 36 months after enrollment 17. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 36 months after enrollment 18. Neurological improvement assessed by mRS at 36 months 19. Combined events such as stroke, myocardial infarction, and vascular death within 36 months after enrollment.
Inclusion Criteria	1. Age 35-80 years 2. Participants with primary or recurrent sICAS (a recent TIA[<90 days] or ischemic stroke [14-90 days] before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial artery) receiving treatment with at least one antithrombotic drug and/or standard medical management of vascular risk factors 3. Diagnosed by DSA: a major intracranial artery (terminal internal carotid artery [ICA] [C4-C7 segments], middle cerebral artery [MCA] M1 segment, vertebral artery [VA] V4 segment, and basilar artery [BA]) severe atherosclerotic stenosis (70-99% according to WASID method) with lesion length $\leq$ 10mm, diameter $\geq$1.5mm, and normal distal artery. (Regarding the curvature and angle of the lesion, whether the patient is enrolled in BASIS will be assessed according to the investigator's determination based on the patient's anatomy) 4. Informed consent signed.

Exclusion Criteria	1. Surgery within the past 30 days or plan to receive major surgery within 90 days 2. Thrombolytic therapy within 24 hours before enrollment 3. Neurological deficits worsen within 24 hours before enrollment 4. Acute ischemic stroke onset within 14 days before enrollment 5. Other intracranial arteries with severe stenosis (70-99%) besides the target artery and its supplying artery 6. Target artery's supplying artery stenosis > 50%. For example, patient with MCA severe stenosis (target artery) and the ipsilateral ICA stenosis > 50% should be excluded; patient with BA severe stenosis (target artery) and the dominant VA stenosis > 50% should be excluded; the patient with extracranial artery (non-lesion side) stenosis > 70% should be excluded; For patients with balanced VA, if bilateral VA stenosis > 70% at the same time, should be excluded (unable to determine which VA is the lesion artery). However, if patient's dominant VA is the lesion artery with a dysplasia or slender contralateral VA or a non-dominant contralateral VA terminating at posteroinferior cerebellar artery, should not be excluded. 7. Participant with perforator stroke (except for severe stenosis of supplying artery combined with hemodynamic compromise or poor collaterals) 8. Baseline mRS ≥ 3 9. Non-atherosclerotic diseases (e.g., arterial dissection, moyamoya disease, vascular inflammatory lesions caused by infection, autoimmune diseases, post-radiation, postpartum status; developmental or genetic abnormalities such as fibromuscular dysplasia, sickle cell anemia, suspected vasospasm) 10. Target artery with severely calcified and adjacent to stenosis 11. Suspected ischemic event caused by embolism or the arterial
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	embolism from extracranial segment (ipsilateral chest or neck vascular occlusive disease) or potential cardioembolism (e.g., atrial fibrillation, mitral stenosis, patent foramen ovale, left ventricular thrombus, myocardial infarction within 6 weeks, etc.) 12. Coexistent intracranial tumor, aneurysm or arteriovenous malformation 13. Intracranial hemorrhage within the past 3 months, including parenchymal hemorrhage, intraventricular hemorrhage, subarachnoid hemorrhage, epidural hemorrhage or subdural hemorrhage, etc. 14. Angioplasty procedure (including balloon dilatation, stenting, or endarterectomy) performed at the original target vessel or its primary supplying artery, or a planned stenting procedure 15. Unable to receive dual antiplatelet therapy due to other diseases 16. Tortuous vascular approach that cannot be stabilized to obtain vascular access 17. Allergy to heparin, aspirin, clopidogrel, contrast agents, anesthetics, and balloon components 18. Hemoglobin < 100g/L, platelet count < 100×10^9/L, international normalized ratio (INR) >1.5 (irreversible), Coagulation dysfunction or uncorrectable bleeding factors 19. Severe liver or kidney dysfunction. ALT > three-fold higher than the upper limit of normal value or AST > three-fold higher than the upper limit of normal value, or serum creatinine > two-fold higher than the upper limit of normal value 20. Pregnant and lactating women 21. Participants with renal artery, radial artery, and coronary artery requiring concurrent intervention 22. Expected survival is less than < 1 year 23. Unable to complete follow-up due to mental illness, cognitive or
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	emotional disorders 24. Participants participating in other drug/medical device clinical trials who have not yet completed the program requirements 25. Participants are not suitable for the BASIS trial per investigator's opinion.
Sample Size	According to previous literature : ①we assume a 7% event rate of the primary endpoint in the experimental group ; ②a 15% event rate of the primary endpoint in the control group ; ③an 80% statistical power at a one-sided α of 2.5% and a 10% attrition rate ; ④the experimental group and the control group are allocated according to a 1:1 ratio. Finally, a total of 512 patients (256 per group) are planned to be enrolled.
Follow-up	All participants will be followed up by the on-site neurologist at baseline, the day of DSA, discharge, 30$\pm$7 days, 90$\pm$7 days, 6 months$\pm$14 days, 1year $\pm$30 days and up to 3years (at 6-month intervals after 1 year).
Study introduction period	After the study is formally started, the participating study centers copy the video (including the DSA and endovascular process) of 3-5 cases of balloon angioplasty in their centers and transfer the video to the core laboratory. After consistency is passed, the group continues to enrollment. This process is to ensure the consistency of the balloon angioplasty procedure with the protocol. The first 100 participants completed the peri-operative (30 days) observation, mainly collecting the adverse events during the peri-operative period. The PI will need to meet with other investigators from the participating centers for interim discussion to evaluate the feasibility of this treatment plan and the continuation of later research, and follow-up research can be conducted later.

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1139 **2. Abbreviations**

ACA	Anterior Cerebral Artery
ACE	Angiotensin Converting Enzyme
ADA	American Diabetes Association
AE	Adverse Event
ARB	Angiotensin Receptor Blocker

BMI	Body Mass Index
BP	Blood Pressure
CEC	Clinical Events Adjudication Committee
CFDA	China Food and Drug Administration
CHANCE	Clopidogrel in High-risk patients with Acute Nondisabling Cerebrovascular Events
CI	Confidence Interval
CRF	Case Report Form
CT	Computer Tomography
CTP	Computed Tomography Perfusion Imaging
DAPT	Dual Antiplatelet Therapy
DICOM	Digital Imaging and Communications in Medicine
DSA	Digital Subtraction Angiography
DSMB	Data Safety Monitoring Board
EoT	End of Trial
EQ-5D	EuroQOL five Dimensions questionnaire
FAS	Full Analysis Set
FFR	Fractional Flow Reserve
HDL	High Density Lipoprotein
ICAD	Intracranial Atherosclerotic Disease
ICAS	Intracranial Atherosclerotic Stenosis
IDL	Intermediate Density Lipoproteins
IEC	Independent Ethics Committee
IRB	Institutional Review Board
IVRS	Inter-voice Response System Inter-voice
LDL	Low Density Lipoprotein
LFT	Liver Function Test
MCA	Middle Cerebral Artery
MI	Myocardial Infarction
MoCA	Cognitive Functional Outcome by the Montreal Cognitive Assessment

mRS	Modified Rankin Scale
NMPA	National Medical Products Administration
PTAS	Percutaneous Transluminal Angioplasty and Stenting
SAE	Serious Adverse Event
SAMMPRIS	Stenting and Aggressive Medical Management for Preventing Recurrent stroke in Intracranial Stenosis
SAP	Statistical Analysis Plan
SAS	Statistical Analysis Software SAS
SNPs	Single Nucleotide Polymorphisms
TIA	Transient Ischemic Attacks
UADE	Unanticipated Adverse Device Effect
VLDL	Very Low Density Lipoproteins
WASID	Warfarin-Aspirin Symptomatic Intracranial Disease
TCD	Transcranial Doppler

3. Schedule of assessments

Visit	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5	Visit 6 Telephone interview	Visit 7	Follow up 1y to 3y (Visit 8,9,10,11)
Assessment	Baseline 15d~0 d	Angiogram 0 d	Discharge	30d±7d	90d±7 d	6 mo±14 d	1y±30 d	6 months intervals
Informed consent	X							
Inclusion/exclusion criteria check	X							
History and physical examination	X							
Vital signs ¹	X	X	X	X	X		X	
Blood laboratory test ² and urine routine test	X		X	X	X		X	
Coagulation function ³	X							
FBG, HbA1c ⁴	X			X	X		X	
Platelet aggregation test and/or CYP219 genotype test ⁵	X							
ECG	X							
MRI/CT	X						X	
CTP ⁶	X ⁹				X		X	
DSA/CTA/MRA	X ⁹						X	
Life-style modification review ⁷	X		X	X	X	X	X	X
mRS	X			X	X	X	X	X
NIHSS (neurological examination)	X		X		X		X	
EQ-5D	X						X	X
MoCA ⁵	X						X	X
HR-VWI,TCD ⁵	X						X	
Medication review and patient compliance survey		X	X	X	X		X	
AE, SAE and endpoints ⁸	X	X	X	X	X	X	X	X ⁸

1. Vital signs: blood pressure, heart rate, respiration rate, body temperature;

2. Blood laboratory test includes blood routine and biochemistry examinations, hepatic and renal function tests.

3. Coagulation function: PT, APTT, TT, FIB, INR;

4. HbA1c is an optional test, but is recommended for patients with Diabetes Mellitus.

5. Platelet aggregation test, CYP219 genotype test, MoCA, HR-VWI and TCD test are optional according to the availability of the test at the center;

6. CTP is an optional test, but is recommended for patients with hemodynamic compromise, poor collateral or perforator stroke.

7. Life-style modification review: including whether to smoke and the number of cigarettes per day; Weight: reflect the patient's weight control through the patient's weight and BMI; Exercise: number of exercises per week and duration of each exercise.

8. After one year of follow-up, follow-up will focus on AE, SAE and endpoint events;

Abbreviations: FBG, fasting blood glucose; HbA1c, Glycosylated Hemoglobin; ECG, electrocardiograph; MRI, magnetic resonance imaging; CT, computerized tomography; CTP, CT perfusion; DSA, digital subtraction angiography; CTA, computerized tomography angiography; MRA, magnetic resonance angiography; digital subtraction angiography (DSA), computerized tomography angiography (CTA), or magnetic resonance angiography (MRA); mRS, modified Rankin scale; EQ-5D, quality-of-life EuroQol-5

1160 Dimensions; MoCA, montreal cognitive assessment; HR-VWI, High-resolution vessel wall imaging; TCD,
1161 transcranial Doppler; AE, adverse event; SAE, serious adverse event.

4. Background

Intracranial atherosclerotic stenosis (ICAS) is a major etiology of stroke worldwide, which is associated with stroke recurrence, substantial morbidity, and mortality, accounting for up to 50% of ischemic strokes in south and east Asia.¹ Despite receiving treatment with aspirin and standard medical management of vascular risk factors, patients with symptomatic ICAS (sICAS) still have as high as 23% 1-year stroke recurrence rate.^{2,3} Therefore, secondary prevention of stroke and reducing mortality in patients with sICAS is a global major issue. Currently, treatment options for sICAS include aggressive medical management (AMM), balloon angioplasty, and stenting (balloon angioplasty combined with stenting); however, the optimal treatment for sICAS remains unclear.

The SAMMPRIS and VISSIT trials have shown high perioperative complication rates and no significant advantage of endovascular stents compared to aggressive medical therapy (AMM) in the long term.^{4,5} However, the CASSISS trial reported that stenting by experienced operators didn't add additional risk of stroke or death within 30 days compared to AMM for sICAS, although it failed to demonstrate a benefit of stenting.⁶ As lower rates of short-term stroke or mortality (peri-procedural or mean follow-up ≤ 3 months) were found in balloon angioplasty than stenting,⁷ balloon angioplasty alone may be a feasible alternative treatment to stenting for sICAS treatment. Moreover, several previous meta-analyses also proved the safety and efficacy of balloon angioplasty for sICAS, which suggested that balloon angioplasty alone may be a potentially promising treatment for sICAS.⁸⁻¹⁰

To date, limited evidence is available from RCTs to determine the efficacy and safety of balloon angioplasty for sICAS patients. Hence, we designed the BASIS (Balloon Angioplasty

for Symptomatic Intracranial Artery Stenosis) randomized controlled trial to explore whether balloon angioplasty plus AMM is superior to AMM alone in patients with sICAS.

5. Objectives and Hypothesis

5.1 Objectives

To investigate whether balloon angioplasty combined with AMM is superior to AMM alone for patients with high-risk sICAS in the primary endpoint (ischemic/haemorrhagic stroke or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization from the qualifying artery beyond 30 days through 12 months after enrollment).

5.2 Hypothesis

Compared to AMM alone, balloon angioplasty combined with AMM could decrease the risk of the primary endpoint by 50% in patients with high-risk sICAS over the 12-month follow-up period.

6. Study Design

Study introduction period

After the study is formally started, the participating study centers copied the video (including the DSA and surgical process) of 3-5 cases of balloon angioplasty in their centers and transfer the video to the core laboratory (Beijing Tiantan Hospital). After the consistency is passed, the group continued to enroll, so as to ensure the consistency of the balloon angioplasty procedure. The annual volume of BA by the neurointerventionalists in the BASIS trial was more than 50 cases. The neurointerventionalists from the participating centers received standard training in technique from the leading center at regular intervals.

The BASIS trial is an investigator-initiated, multicenter, prospective, randomized, open-label, blinded end-point (PROBE) trial that plans to enroll patients with sICAS (defined as a recent TIA (<90 days) or ischemic stroke (14-90 days) attributed to a 70-99% atherosclerotic stenosis of a major intracranial artery including terminal internal carotid artery [ICA] C4-C7 segment, middle cerebral artery [MCA] M1 segment, vertebral artery [VA] V4 segment and basilar artery [BA]) determined by DSA. The patients are randomly assigned to either the experimental group or the control group at a 1:1 ratio.

Experimental group: patients will receive balloon angioplasty (details in procedure part) plus AMM (same as the control group);

Control group: patients will receive the AMM alone.

✓ Drug administration plan of AMM: aspirin 100 mg daily for the entire follow-up and clopidogrel 75mg daily for the first 90 days after enrollment. Whether to continue taking clopidogrel after 90 days will be assessed according to the condition of the patient, clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance; atorvastatin were used 20mg-80mg/day for one year after enrollment; the dose was adjusted to control LDL <1.8mmol/L or <70mg/dl. Whether to continue taking atorvastatin after one year will be assessed according to the patient's condition.

✓ Risk factor management—Control blood pressure 130-140/90-100 mmHg and LDL <1.8mmol/L or <70mg/dl

The first 100 participants completed the peri-operative (30 days) observation, mainly collecting the adverse events during the peri-operative period. The PI needs to meet with other investigators from the participating centers for interim discussion to evaluate the feasibility of

this treatment plan and the continuation of later research, and follow-up research can be conducted later.

7. Study Endpoint

7.1 Primary endpoint

The primary endpoint is stroke or death within 30 days after enrollment, or any ischemic stroke or revascularization of the qualifying artery beyond 30 days through 12 months after enrollment.

Ischemic stroke is defined as a new focal neurologic deficit of sudden onset, which is confirmed on brain CT or MRI. Symptomatic brain hemorrhage is defined as parenchymal, subarachnoid, or intraventricular hemorrhage detected on CT or MRI, which is associated with new neurologic symptoms (headache, change in the level of consciousness, or focal neurologic symptoms), lasting over 24 hours or a seizure. Symptomatic brain hemorrhage will be considered a primary endpoint only when it occurs within 30 days after enrollment or within 30 days after the balloon angioplasty procedure during follow-up. Revascularization of the culprit artery will be considered a primary endpoint if it occurs from 30 days through 12 months after enrollment and fulfills one of the following requirements:

c. Acute revascularization: acute qualifying artery occlusion accompanied by neurological deficit, requiring intravenous thrombolysis, intraarterial thrombolysis, mechanical thrombectomy, balloon/stent angioplasty, or intracranial-extracranial bypass.

d. Elective revascularization: neurologic symptom-driven revascularization, including BA, stent implantation, or intracranial-extracranial bypass, if the participant fulfills one of the following conditions:

- 3) Ischemic stroke caused by the culprit artery stenosis: a new focal neurological deficit of sudden onset attributed to the territory of the culprit artery, which is confirmed as a recurrent stroke on brain CT or MRI (follow-up imaging will be compared to baseline imaging for the detection of new lesions)
- 4) Culprit artery stenosis that causes recurrent transient ischemic attack lasting longer than 10 minutes or new disabling neurological symptoms (paroxysmal limb weakness/numbness, inarticulateness, diplopia, or ataxia) compared to the baseline qualifying symptoms.

7.2 Secondary endpoints

1. Any stroke (ischemic or hemorrhage stroke) or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up
2. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 90 days after enrollment
3. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 90 days after enrollment
4. mRS at 90 days
5. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause death within 12 months after enrollment
6. Revascularization of the target artery within 12 months after enrollment
7. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 12 months after enrollment
8. mRS at 12 months

- 1274 9. Restenosis of the target artery within 12 months (defined as stenosis >70% or increased by
1275 30% based on following-up neurovascular imaging)
- 1276 10. Combined events such as stroke, myocardial infarction, or vascular death within 12 months
1277 after enrollment
- 1278 11. Quality of life assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire) within 12
1279 months after enrollment
- 1280 12. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause
1281 death within 24 months after enrollment
- 1282 13. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery
1283 within 24 months after enrollment
- 1284 14. mRS at 24 months
- 1285 15. Combined events such as stroke, myocardial infarction, or vascular death within 24 months
1286 after enrollment
- 1287 16. Any stroke (ischemic or hemorrhage stroke) in the territory of the target artery or all-cause
1288 death within 36 months after enrollment
- 1289 17. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery
1290 within 36 months after enrollment
- 1291 18. Neurological improvement assessed by mRS at 36 months
- 1292 19. Combined events such as stroke, myocardial infarction, or vascular death within 36 months
1293 after enrollment.

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1295 **8. Inclusion and Exclusion Criteria**

8.1 Inclusion criteria

1. Age 35-80 years

5. Participants with primary or recurrent sICAS (a recent TIA [<90 days] or ischemic stroke [14-90 days] before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial artery) receiving treatment with at least one antithrombotic drug and/or standard medical management of vascular risk factors

6. Diagnosed by DSA: a major intracranial artery (terminal internal carotid artery [ICA] [C4-C7 segments], middle cerebral artery [MCA] M1 segment, vertebral artery [VA] V4 segment, and basilar artery [BA]) severe atherosclerotic stenosis (70-99% according to WASID method¹¹) with lesion length ≤ 10 mm, diameter ≥ 1.5 mm, and normal distal artery. (Regarding the curvature and angle of the lesion, whether the patient is enrolled in BASIS will be assessed according to the investigator's determination based on the patient's anatomy)

7. Informed consent signed.

8.2 Exclusion criteria

1. Surgery within the past 30 days or plan to receive ≥ 3 -grade surgery within 90 days

2. Thrombolytic therapy within 24 hours before enrollment

3. Neurological deficits worsen within 24 hours before enrollment

4. Acute ischemic stroke onset within 14 days before enrollment

5. Other intracranial arteries with severe stenosis (70-99%) besides the target artery and its supplying artery

6. Target artery's supplying artery stenosis $> 50\%$. For example, patient with MCA severe stenosis (target artery) and the ipsilateral ICA stenosis $> 50\%$ should be excluded; patient

with BA severe stenosis (target artery) and the dominant VA stenosis > 50% should be excluded; the patient with extracranial artery (non-lesion side) stenosis > 70% should be excluded; For patients with balanced VA, if bilateral VA stenosis > 70% is present at the same time, the patient should be excluded (unable to determine which VA is the lesion artery). However, if the patient's dominant VA is the lesion artery with a dysplasia or slender contralateral VA or a non-dominant contralateral VA terminating at the posteroinferior cerebellar artery, the patient should not be excluded.

7. Participant with perforator stroke (except for severe stenosis of supplying artery combined with hemodynamic compromise or poor collaterals)¹²

8. Baseline modified Rankin Scale ≥ 3

9. Non-atherosclerotic diseases (e.g., arterial dissection, moyamoya disease, vascular inflammatory lesions caused by infection, autoimmune diseases, post-radiation, postpartum status; developmental or genetic abnormalities such as fibromuscular dysplasia, sickle cell anemia, suspected vasospasm)

10. Target artery with severe calcification

11. Suspected ischemic event caused by embolism or arterial embolism from an extracranial segment (ipsilateral chest or neck vascular occlusive disease) or potential cardioembolism (e.g., atrial fibrillation, mitral stenosis, patent foramen ovale, left ventricular thrombus, myocardial infarction within 6 weeks, etc.)

12. Coexistent intracranial tumor, aneurysm or arteriovenous malformation

- 1338 13. Intracranial hemorrhage within the past 3 months, including parenchymal hemorrhage,
1339 intraventricular hemorrhage, subarachnoid hemorrhage, epidural hemorrhage or subdural
1340 hemorrhage, etc.
- 1341 14. Angioplasty procedure (including balloon dilatation, stenting, or endarterectomy) already
1342 performed at the original target vessel or its primary supplying artery, or a planned stenting
1343 procedure
- 1344 15. Unable to receive dual antiplatelet therapy due to other diseases
- 1345 16. Tortuous vascular approach not amenable to vascular access
- 1346 17. Allergy to heparin, aspirin, clopidogrel, contrast agents, anesthetics, and balloon components
- 1347 18. Hemoglobin < 100g/L, platelet count < 100×10^9 /L, international normalized ratio
1348 (INR) >1.5 (irreversible), Coagulation dysfunction or uncorrectable bleeding factors
- 1349 19. Severe liver or kidney dysfunction. ALT > three-fold higher than the upper limit of normal
1350 value or AST > three-fold higher than the upper limit of normal value, or serum creatinine >
1351 two-fold higher than the upper limit of normal value
- 1352 20. Pregnant and lactating women
- 1353 21. Participants with renal artery, radial artery, and coronary artery requiring concurrent
1354 intervention
- 1355 22. Expected survival is less than < 1 year
- 1356 23. Unable to complete follow-up due to mental illness, cognitive or emotional disorders
- 1357 24. Participants participating in other drug/medical device clinical trials who have not yet
1358 completed the program requirements
- 1359 25. Participants are not suitable for BASIS trial per investigator's opinion.

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1361 **9. Informed Consent**

1362 The clinical trial must follow the Helsinki Declaration and the relevant clinical trial norms and
1363 regulations in China. Informed consent was designed based on NMPA regulations and ICH
1364 guidelines. Each patient who agreed to participate the study will be invited to sign an informed
1365 consent form. Before signing the informed consent form, a specially trained researcher
1366 (understanding the key points of the study and the related risks of the study, the confidentiality
1367 of the management records and the source of funds, etc.) will introduce the research purpose,
1368 study plan, study cycle, potential risks and expected benefits, cost and compensation, voluntary
1369 participation principle, treatment principle of study related risks, etc.

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1371 **10. Randomization**

1372 The BASIS trial used an interactive web response system (IWRS) for central randomization
1373 stratified by study centers. Patients who sign the informed consent and meet all inclusion criteria
1374 and do not meet any exclusion criteria will be enrolled in the study. After entering the eligible
1375 patient's necessary information into the web-based system, the researcher in each center will
1376 obtain a random code as well as the corresponding group allocation information from the IWRS.
1377 Eligible patients will be randomized into either the experimental group or the control group at a
1378 1:1 ratio (**Figure 1**). Finally, researchers and research assistants need to print randomization
1379 application forms and center randomization lists.

The researcher team should archive the randomized list and related files (randomized application form) together in the central researcher folder (archived by the clinical study department of the center).

11. DSA evaluation and lesion evaluation

In order to ensure the quality of DSA images in the study, we will choose a clinical research center with DSA facilities. Before the start of the study, researchers from all centers will participate in the training of image generation, collection and evaluation, and obtain corresponding qualifications.

When the patient is enrolled, the central laboratory will obtain the CD disc of the patient's MRI / HRMRI imaging, CT perfusion imaging, and DSA before balloon angioplasty for central review. DSA imaging needs to show the residual vessel diameter at the maximum stenosis and the maximum length of the stenosis. If the patient underwent balloon angioplasty, arterial imaging after angioplasty should also be sent to the central laboratory for central review. The standard of technical success is the residual stenosis of the target vessel $\leq 50\%$.

12. Balloon Angioplasty

Antithrombotic protocol

Pre-procedure: All patients who are scheduled to undergo balloon angioplasty should receive dual antiplatelet therapy (aspirin 100 mg per day and clopidogrel 75 mg per day) for at least three days before the procedure.

Intra-procedure: Intravenous anticoagulant such as heparin during the procedure.

Post-procedure: Aspirin 100 mg per day must be used for the entire duration of follow-up, and clopidogrel 75 mg per day or ticagrelor 90mg twice a day must be used for 90 days after randomization. After 90 days, whether to continue to take it will be assessed according to the patient's condition and the researcher's opinion. Atorvastatin will be taken 20mg-80mg/day for one year after enrollment; Whether to continue taking atorvastatin after one year will be assessed according to the patient's condition.

Anesthesia strategy and arterial access

The procedure should be performed under general anesthesia. Femoral artery access is recommended (radial artery access is allowed for patients with tortuosity of the aortic arch, or aorta).

Procedure steps

8. A stable vascular access with a long sheath or guiding catheter (an intermediate catheter is recommended for participants with tortuous access), which should be placed as distal as possible, is suggested to provide adequate support for performing the balloon angioplasty. Collateral status assessment based on a whole brain DSA is recommended before balloon angioplasty according to the ASITN/SIR collateral flow grading system (poor collateral: ASITN/SIR < 3).¹³
9. An 0.014-inch microwire with a microcatheter (or not) passes through the lesion to reach the distal branch of the target artery.
10. A balloon (recommending the Neuro RX and Neuro LPS Intracranial Balloon Dilation Catheter [Sinomed Inc., Tianjin, China]) is navigated over the microwire to the lesion of the target artery.

Balloon size selection: the balloon length should cover the lesion completely and cover at least 2 mm beyond each end of the lesion. The balloon diameter is determined according to the physician's opinion (submaximal angioplasty is suggested: balloon with a diameter no more than 70% of the proximal artery diameter).

11. The balloon should be slowly inflated to the nominal pressure, and maintained for 10-30 seconds before deflating slowly. This step can be repeated 2-3 times if the dilation effect of balloon angioplasty is not satisfactory. For patients with extremely severe stenosis, a relatively smaller balloon can be used for pre-dilation.

12. Technical success: residual stenosis $\leq 50\%$ of the proximal artery diameter with modified Thrombolysis in Cerebral Infarction (mTICI)¹⁴ of 3 grade without arterial dissection that impairs the distal blood flow.

13. Rescue stent implantation is allowed when: (1) residual stenosis $\geq 70\%$ after balloon dilation or the antegrade flow is unstable (mTICI $< 2b$); (2) arterial dissection impairing the distal blood flow (mTICI $< 2b$); (3) development of thrombosis or embolization that jeopardizes distal perfusion. Other measures such as intravenous/intra-arterial thrombolysis, infusion of GP IIb/IIIa inhibitors, are based on the physician's experience, which can be used alone or in combination with rescue stenting.

14. A final DSA should be performed after 10–15 minutes of observation.

13. Aggressive medical management

13.1 Antiplatelet treatment

AMM will be identical in both arms. All the necessary blood and imaging tests are collected in the screening period before randomization for the following management of risk factors. Both arms are prescribed with an AMM regime (includes aspirin 100 mg per day for the entire follow-up and clopidogrel 75mg per day for the first 90 days after enrollment). For the record: clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance (Platelet aggregation rate of adenosine diphosphate >40% or Loss-of-function alleles of CYP2C19 is detected).

13.2 Risk factor management

Neurologists and study investigators at each site will be responsible for the management of patient's risk factors. They will receive training on risk factor management according to the Chinese Stroke Association guidelines for the clinical management of cerebrovascular disorders.¹⁵ All study investigators are regularly trained according to the BASIS protocol every 6 months. If a patient fails to reach the target, a face-to-face follow-up with the local neurologist to modify their medical regime will be necessary.

13.2.1 Primary risk factors

Achieving target LDL cholesterol

The baseline serum low-density lipoprotein cholesterol (LDL-c) level is recorded before enrollment. The target LDL-c level is < 1.8mmol/L or 70 mg/dl. If the patient's LDL level is higher than the target level, he/she should modify the lipid-lowering drugs. Liver enzyme (AST / ALT) levels will be measured at the beginning of the study and at each visit point.

Blood pressure management

The blood pressure will be checked at screening, randomization, discharge, 30 days, 90 days, 180 days, 1 year, and the end of the trial. Patients may need to visit local sites for blood pressure measurement and modify their medical regime if necessary. The target blood pressure is no more than 140mmHg/90mmHg for patients without DM and no more than 130mmHg/80mmHg for patients with DM. Antihypertensive drugs include Angiotensin-converting enzyme inhibitor (ACEI), Angiotensin Receptor Blocker (ARB), beta-blocker, calcium channel antagonist (CCB), diuretic, vasodilator, etc.

13.2.2 Secondary risk factors

Non-HDL cholesterol

The target level of non-HDL cholesterol is < 100 mg/dL. Non-HDL cholesterol includes low-density lipoproteins (LDL), very-low-density lipoproteins (VLDL), and intermediate-density lipoproteins (IDL). When a larger statin dosage fails to lower the LDL level, or if the non-HDL cholesterol $\geq$ 100 mg/dL and triglycerides $\geq$ 200mg/dL, then other lipid-lowering drugs in addition to statin are recommended, such as ezetimibe or PCSK9 inhibitors.

Diabetes management

The target of DM management is to achieve HbA1c < 7.0%. Fasting plasma glucose (FPG) and HbA1c levels will be measured at baseline, 30 days, 90 days, and 1 year.

Lifestyle modification

Smoking cessation: investigators at each site will evaluate the patient's smoking status at each follow-up and encourage all subjects to quit smoking as soon as possible.

1487 **Weight management:** investigators at each site will evaluate the weight according to body
1488 mass index (BMI).

1489 **Activity level:** activity level for each patient is also assessed at each follow-up by trained
1490 investigators or coordinators, and moderate exercise of 30 minutes a day, three times per week is
1491 strongly recommended to all patients with athletic ability.

1492 **13.2.3 Research drugs**

1493 During the screening period of BASIS, clopidogrel will be provided free of charge for 3-5 days.

1494 After 90 days of enrollment, clopidogrel will also be provided free of charge. From enrollment
1495 to 1 year, atorvastatin required by patients will be provided free of charge, and other drugs will
1496 be purchased by the patients.

1497 At each visit point, the patient's compliance with treatment will be evaluated by counting the
1498 number of clopidogrel and atorvastatin.

1499 Patients with LDL > 1.8 mmol / L or > 70 mg / dL will be treated with atorvastatin. Whether to
1500 continue taking atorvastatin after 1 year will be assessed according to the patient's condition. If
1501 patients are taking other lipid-lowering drugs, patients are encouraged to switch to atorvastatin
1502 (not mandatory). For patients whose blood pressure does not reach the treatment goal,
1503 antihypertensive drugs will be given. Patients can choose to use but are not limited to the
1504 following antihypertensive drugs:

- 1505 ✓ Diuretic (Chlorthione 25mg)
- 1506 ✓ Angiotensin-converting enzyme inhibitor (ACEI) (Lisinopril 10mg or 40 mg)
- 1507 ✓ Angiotensin Receptor Blocker (ARB) (Candesartan 16mg or 32 mg),
- 1508 ✓ Beta-blocker (Atenolol 50mg or 100mg)

1509 ✓ Calcium channel antagonist (CCB) (Felodipine 5mg or 10 mg)

1510 ✓ Central α agonist (clonidine 0.1mg)

1511 ✓ Vasodilator (hydralazine 50mg)

1512

1513 **14. Patient and Researcher Benefit**

1514 **Patient benefit:** Patients participating in this study will receive more systematic treatment and
1515 follow-up. Clopidogrel is provided free of charge during the screening period and 90 days of
1516 enrollment, and atorvastatin is provided free of charge from enrollment to 1 year. DSA is free at
1517 1-year follow-up. However, other related examinations and medications that need to be
1518 explained are not free, and the balloons used when they are randomly assigned to the
1519 experimental group are not free.

1520 **Researchers benefit:** Researchers participating in this study have the right to give priority to the
1521 use of scientific data to publish and sign scientific research work.

1522

1523 **15. Follow-up**

1524 All patients will be followed up to the primary endpoint of the study or the end of the study.
1525 Unless the patient does not agree to continue participating in the study, efforts will be made to
1526 obtain complete endpoint information (whether the patient is cured or dead). If the patient
1527 discontinued this study and received other treatments, such as stenting, but agreed to provide
1528 endpoint information, follow-up will continue.

1529 All participants will be followed up by the on-site neurologists at baseline, the day of DSA,
1530 discharge, 30 ± 7 days, 90 ± 7 days, $6 \text{ months} \pm 14$ days, $1 \text{ year} \pm 30$ days, and up to 3 years (at 6

1531 months intervals after 1 year). Follow-up visits will be conducted by telephone at 6-month and
1532 post-1-year periods and will be evaluated in person at other visits. At each follow-up visit, the
1533 participants' medications, laboratory tests, risk factor management (as described above), and
1534 possible adverse events/endpoints are reviewed by an experienced neurologist and/or
1535 neurointerventionalist. All patients are required to undergo a neurovascular imaging
1536 examination including DSA, CTA or MRA at 1-year follow-up.

1537 **Visit 1: Screening**

- 1538 ✓ Signed informed consent
- 1539 ✓ Review inclusion/exclusion criteria
- 1540 ✓ Randomization-experimental group: balloon angioplasty+AMM; control group: AMM
- 1541 ✓ Medical history, concomitant medication, demographic data collection
- 1542 ✓ Ischemic stroke history record
- 1543 ✓ Vital signs, blood routine, urine routine, blood biochemistry, coagulation function
1544 examination
- 1545 Vital signs: Blood pressure, heart rate, respiration, body temperature;
- 1546 blood biochemistry: Renal function: creatinine, urea (or urea nitrogen), glucose, uric acid,
1547 total bilirubin, direct bilirubin, indirect bilirubin, lactate dehydrogenase, creatine kinase
1548 isoenzyme; liver function: alkaline phosphatase, glutamyl transpeptidase, aspartate
1549 aminotransferase, alanine aminotransferase; blood lipids: total cholesterol, low-density
1550 lipoprotein, high-density lipoprotein, triglyceride;
- 1551 Coagulation function: PT, APTT, TT, FIB, INR
- 1552 ✓ Fasting blood glucose, HbA1c (optional)

- 1553 ✓ Pregnancy test
- 1554 ✓ Electrocardiogram
- 1555 ✓ DSA
- 1556 ✓ MRI/HRMRI (HRMRI is unnecessary; if MRI is not available, CT can replace MRI)
- 1557 ✓ CTP (anterior circulation sICAS)
- 1558 ✓ Patients' smoking cessation, weight, and exercise data are collected. Smoking cessation:
1559 including whether to smoke and the number of daily cigarettes; weight: use the patient's
1560 weight and BMI to reflect the patient's weight control; exercise: weekly exercise frequency,
1561 each exercise duration
- 1562 ✓ mRS, NIHSS, MoCA, EQ-5D (MoCA is unnecessary)
- 1563 ✓ Platelet function test and/or CYP219 genotype test according to the availability of the
1564 center, this item is unnecessary
- 1565 ✓ TCD (recommend but not necessary)
- 1566 ✓ Research drug distribution: Clopidogrel (only for patients who are enrolled in BASIS)
- 1567 ✓ Aspirin and clopidogrel medication orders
- 1568 ✓ Concomitant medication collection
- 1569 ✓ AE, SAE and endpoint observation
- 1570 **Visit 2: Treatment**
- 1571 ✓ Treatment according to the randomization scheme
- 1572 ✓ Concomitant medication collection
- 1573 ✓ Vital signs

- 1574 ✓ Research drug distribution and recovery (Randomly successful patients are distributed 30
1575 days follow-up research drug; Randomly failed patients recovered research drugs issued
1576 during the screening period; randomly successful patients are given clopidogrel and
1577 atorvastatin)
- 1578 ✓ Aspirin, clopidogrel and atorvastatin medication orders
- 1579 ✓ Patient compliance survey
- 1580 ✓ AE, SAE and endpoint observation
- 1581 **Visit 3: Discharge**
- 1582 ✓ Concomitant medication collection
- 1583 ✓ Vital signs
- 1584 ✓ Blood routine, urine routine, blood biochemistry (optional)
- 1585 ✓ NIHSS
- 1586 ✓ Patients' smoking cessation, weight, and exercise data are collected
- 1587 ✓ Patient compliance survey
- 1588 ✓ The disease treatment-related cost information collection
- 1589 ✓ AE, SAE and endpoint observation
- 1590 **Visit 4: 30±7 days after enrollment**
- 1591 ✓ Concomitant medication collection
- 1592 ✓ Vital signs, blood routine, urine routine, blood biochemistry
- 1593 ✓ Fasting blood glucose, HbA1c (optional)
- 1594 ✓ Patients' smoking cessation, weight, and exercise are collected
- 1595 ✓ mRS

- 1596 ✓ Research drug distribution: clopidogrel and atorvastatin
- 1597 ✓ Aspirin, clopidogrel and atorvastatin calcium orders
- 1598 ✓ Patient compliance survey
- 1599 ✓ The disease treatment-related cost information collection
- 1600 ✓ AE, SAE and endpoint observation
- 1601 **Visit 5: 90±7 days after enrollment**
- 1602 ✓ Concomitant medication collection
- 1603 ✓ Vital signs, blood routine, urine routine, blood biochemistry
- 1604 ✓ Fasting blood glucose, HbA1c (optional)
- 1605 ✓ CTP (optional)
- 1606 ✓ Patients' smoking cessation, weight, and exercise data are collected
- 1607 ✓ mRS and NIHSS
- 1608 ✓ Research drug distribution: atorvastatin
- 1609 ✓ Aspirin and atorvastatin medication orders
- 1610 ✓ Patient compliance survey
- 1611 ✓ The disease treatment-related cost information collection
- 1612 ✓ AE, SAE and endpoint observation
- 1613 **Visit 6: 6 months±14 days after enrollment (interview by telephone)**
- 1614 ✓ Concomitant medication collection
- 1615 ✓ Patients' smoking cessation, weight, and exercise data are collected
- 1616 ✓ mRS
- 1617 ✓ AE, SAE and endpoint observation

1618 **Visit 7: 1 year±30 days after enrollment**

- 1619 ✓ Concomitant medication collection
- 1620 ✓ Vital signs, blood routine, urine routine, blood biochemistry
- 1621 ✓ Fasting blood glucose, HbA1c (optional)
- 1622 ✓ MRI/HRMRI (HRMRI is unnecessary; if MRI is not available, CT can replace MRI)
- 1623 ✓ CTP, DSA (DSA is necessary and CTP is unnecessary)
- 1624 ✓ Patients' smoking cessation, weight, and exercise data are collected
- 1625 ✓ mRS, NIHSS, MoCA, EQ-5D (MoCA is unnecessary)
- 1626 ✓ TCD (recommend but not necessary)
- 1627 ✓ Research drug recovery: atorvastatin
- 1628 ✓ Patient compliance survey
- 1629 ✓ Follow-up aspirin medication orders
- 1630 ✓ The disease treatment-related cost information collection
- 1631 ✓ AE, SAE and endpoint observation

1632 **Follow-up after 1 year: every 6 months ± 30 days after 1-year enrollment**

- 1633 Follow-up every 6 months (visit 8, visit 9, visit 10, visit 11, visit 9, and visit 11 require patients
- 1634 to undergo face-to-face follow-up).
- 1635 Visit by telephone:
- 1636 ✓ Patients' smoking cessation, weight, and exercise are collected
- 1637 ✓ Follow-up aspirin medication
- 1638 ✓ AE, SAE and endpoint observation
- 1639 Face-to-face visit:

1640 ✓ Patients' smoking cessation, weight, and exercise data are collected

1641 ✓ mRS, MoCA, EQ-5D (MoCA is unnecessary)

1642 ✓ Follow-up aspirin and atorvastatin medication orders

1643 ✓ AE, SAE and endpoint observation.

1644 **16. Management of (transient) ischemic events during follow-up**

1645 If any patient in each group had potential clinical neurological events (transient cerebral
1646 ischemia, ischemic stroke), neurologists would determine whether to perform a vessel imaging
1647 examination, including DSA. The most effective treatment for ICAS patients with transient
1648 cerebral ischemia or ischemic stroke is unknown. The researchers determine the best treatment
1649 according to the patient's condition and actively control the risk factors.

1650 If patients in the experimental group experienced transient cerebral ischemia or ischemic
1651 stroke, accompanied by restenosis $\geq 70\%$ diagnosed by DSA, investigators composed of
1652 neurologists and neurointerventional experts will give advice on whether they should next
1653 undergo revascularization or medical management alone. Revascularization is not recommended
1654 for patients with restenosis $< 70\%$.

1655 **17. Assessment of endpoint and adverse events (AEs)**

1656 **17.1 Endpoint event assessment**

1657 The endpoint events include target vessel haemorrhagic stroke, ischemic stroke, death, and
1658 target vessel revascularization. If a patient has a potential primary endpoint event, the researcher
1659 must report to a third-party monitoring company project team within 48 hours, including the
1660 patient's complete medical records. The project team will submit the patient's data to the event
1661 arbitration committee, and the event arbitration committee will complete the arbitration within 7

days. The endpoint event arbitration committee is composed of three members. First, two members use back-to-back arbitration. If the evaluation opinions are inconsistent, a third evaluation expert will cast a decisive vote.

17.2 AEs assessment

An AE is the presence of any unexpected medical condition during or after being treated with medical devices. It includes symptoms, signs, or abnormal laboratory parameters that can be unrelated to treatment. An AE that fulfills at least one of the following criteria is classified as a severe AE(SAE): AE that results in death; is life-threatening; requires/prolongs inpatient hospitalization; results in serious disability; or requires medical intervention to prevent one of the above-mentioned outcomes. If a potential endpoint occurs, the committee board meeting will be convened to evaluate whether such an event can be categorized as the primary endpoint.

Unanticipated adverse device effect (UADE): Any serious adverse reaction affecting health or safety, or any life-threatening event or death related to (caused by) the device. The nature, severity, or degree of such serious adverse reactions, life-threatening events or deaths, or other unexpected device-related serious events involving patient rights, safety, or welfare, were not previously identified in the study plan.

The following events are considered as possible (expected) complications of balloon angioplasty:

Vasospasm	Neurological symptoms	Coagulation disorders
Dissection	Pseudoaneurysm	Death
Vessel occlusion	Renal insufficiency/failure	embolism (air, tissue, thrombus or foreign body)
Vessel perforation	Respiratory distress/failure hematoma	Pain, and/or puncture site infection

Vessel rupture	Seizure	Hemorrhage
Vessel embolism	Infection	Hypertension
Stroke	Drug reaction of antiplatelets	Hypotension
Transient ischemic attack	Syncope	

1680

1681 **17.3 SAE record and report**

1682 The collection time of adverse events and serious adverse events in the study begins with
 1683 signing the informed consent, across the study period, until the end of the study. SAEs of
 1684 patients who discontinued treatment in advance should also be recorded.

1685 When any SAE occurs during the study, the researcher should immediately treat appropriately
 1686 for the patient, and report in writing within 24 hours to the clinical trial institution, ethics
 1687 committee, and sponsor (Beijing Tiantan Hospital, Capital Medical University). For death
 1688 events, clinical trial institutions and researchers should provide the ethics committee and
 1689 sponsors (Beijing Tiantan Hospital, Capital Medical University) with all the required
 1690 information.

1691 **17.4 Prediction of AEs and countermeasures**

1692 AEs that may occur during the procedures include, but are not limited to:

1693 **Thromboembolic events**

1694 The possible causes include no anticoagulation or insufficient anticoagulation, insufficient
 1695 antiplatelet aggregation treatment before and after the procedure or aspirin resistance.
 1696 Anticoagulation and antiplatelet aggregation should be standardized. If embolization occurs in
 1697 the important vessels, it should be handled by a professional doctor per the patient's condition.

1698 **Cerebral vasospasm**

1699 Cerebral vasospasm can be caused by subarachnoid hemorrhage (SAH), or by stimulation of
1700 endovascular catheters and guidewires. If vasospasm occurs, the professional doctor will give
1701 symptomatic treatment per the patient's condition.

1702 **Acute vessel occlusion**

1703 Possible causes: intimal injury, platelet aggregation, thrombosis, etc., usually occur within 30
1704 minutes after balloon angioplasty or are related to intimal avulsion. Preventive measures:
1705 including aspirin and ticlid or clopidogrel antiplatelet drugs can effectively reduce acute
1706 thrombosis events. Under strict preprocedure antiplatelet therapy and intraprocedure
1707 anticoagulation, acute vessel occlusion events can be avoided.

1708 Treatment measures: Once acute vessel occlusion occurs, reasonable treatment should be
1709 performed immediately for the cause of occlusion. Thrombolysis can be performed in patients
1710 with acute thrombosis.

1711 **Thrombosis and plaque shedding**

1712 It is mainly due to plaque fragmentation and shedding caused by squeezing plaque during
1713 balloon dilatation, and sometimes thrombosis caused by catheter and guide wire. The use of
1714 brain protection devices and intraprocedure anticoagulation are the basic measures to prevent
1715 this event.

1716 **Vessel rupture**

1717 Such complications can directly lead to death. When balloon angioplasty is performed at some
1718 sites, the selected balloon is preferably slightly smaller than the diameter of the target vessel.

Once vessel rupture occurs, symptomatic treatment should be performed immediately by a professional doctor.

Cerebral hyperperfusion syndrome

For severe stenotic lesions, the autoregulation ability of distal vessels decreases due to long-term ischemia. The sudden expansion of the lesion site significantly increases the blood flow. The distal vessels, especially the cortical arteries, cannot immediately adapt to this sudden increase in blood flow and passively expand, resulting in cerebral hyperperfusion syndrome. The main clinical presentations include headache, nausea, vomiting, epilepsy, and disturbance of consciousness. Some patients can suffer intracranial hemorrhage. The possibility of this occurrence should be fully recognized before the procedure, and blood pressure should be appropriately reduced during and after the procedure to prevent this event. If postoperative cerebral hyperperfusion syndrome occurs, symptomatic treatment should be performed according to the patient's condition (such as antihypertensives, analgesics, antiepileptic drugs, hyperosmolar agent to reduce intracranial pressure, surgical treatment of cerebral hemorrhage, etc.).

18. Core Laboratory

Imaging will be adjudicated by an independent imaging core lab (China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence). All of the blood samples and imaging data will be collected, transported, and preserved according to the study protocol. Imaging will be blind (clinical features and treatment options for patients), reviewed by experienced MRI experts and DSA experts (over 10-year working experience).

1741

1742 **19. Quality control**

1743 To ensure that the collected data is concordant with the protocol, each sub-center of the BASIS
1744 trial will be regularly scrutinized, and by comparing data in the Electronic Data Capture System
1745 (EDC) and data in the original documents (case report form [CRF] vs. source documents) to
1746 confirm data consistency. If a patient develops an AE/severe AE or completed a 1-year
1747 observation after receiving treatment from non-study sites, duplicate copies of the medical
1748 documents will be collected for future reviews.

1749

1750 **20. Statistical Analysis**

1751 The composite event rates and corresponding 95% CIs of the primary outcome in the two
1752 treatment groups will be estimated by Kaplan-Meier survival analysis and compared by log-rank
1753 test. Hazard ratio between the two groups and its 95% CI will be calculated using a Cox
1754 proportional hazards regression model. Time-to-event endpoints of the secondary outcomes will
1755 also be analyzed using Kaplan-Meier survival analysis and Cox regression, and common odds
1756 ratios of mRS will be estimated using ordinal logistic regression. The widths of the intervals will
1757 not be adjusted for multiplicity for secondary outcomes. The main analysis of this study will be
1758 conducted based on the intention-to-treat principle, and per protocol analysis and as treated
1759 analysis will also be conducted as sensitivity analyses. All analyses will be performed using
1760 SAS software, Version 9.4 (SAS Institute). Detailed statistical methods, missing data imputation
1761 and subgroup analyses are described in the statistical analysis plan.

1762 **20.1 Sample size**

According to the VISSIT trial and a previous randomized trial in China, the composite event rate of the primary outcome in the medical group is anticipated to be 15%.^{4,16} For the balloon angioplasty group, we assume a 7% rate of the primary outcome based on studies of angioplasty without stenting^{9,17} and the investigator's clinical practice experience in China. As a result, the sample size will require detection of an absolute difference of 8% between the two groups. A total of 512 patients (256 per group) will be enrolled considering an 80% statistical power at a one-sided α of 2.5% and a 10% dropout rate.

20.2 Subgroup analysis

Subgroup analyses on the primary outcome will be performed in the following subgroups:

- ✓ Age, <65 years old vs. ≥ 65 years old
- ✓ Sex, men vs. women
- ✓ Hypertension, yes vs. no
- ✓ Diabetes, yes vs. no
- ✓ Smoking, yes vs. no
- ✓ Baseline renal function, eGFR <60 ml/min/1.73m² vs. eGFR ≥ 60 ml/min/1.73m²
- ✓ Target vessel stenosis degree, <80% vs. $\geq 80\%$
- ✓ BMI, <25 Kg/m² vs. 25-30 Kg/m² vs. ≥ 30 Kg/m²
- ✓ Hypoperfusion, yes vs. no
- ✓ Lesion location, anterior circulation vs. posterior circulation
- ✓ The mechanism, ischemic stroke vs. TIA

21. Data Safety and Monitoring Board

The data safety and monitoring board (DSMB) is comprised of academic members and an independent statistician. The DSMB is scheduled to have meetings annually to review the study progress to guarantee the trial is in accordance with the standards of ethics and to ensure the safety of all enrolled patients. After each DSMB meeting, the DSMB members will generate a report including all recommendations, and hand it to the steering committee immediately after the meeting.

22. Study organization

Twice a year, the steering committee will meet to oversee the trial and provide strategic guidance. Periodically, the clinical research team of the leading center will meet online with the project team of the clinical research organization to review the trial progress and data monitoring each week. An independent clinical events adjudication committee (CEC) will ensure that defined outcomes are reported and judged uniformly using the same definition by experts who are blinded to the treatment status. (Table 1)

Table 1. Collaboration team of trial and Study Committee

Collaboration team	Unit or Hospital name	Investigator Name
Sponsor	Beijing Tian Tan Hospital, Capital Medical University	NA
CRO	Beijing Jingchengtong Medicine Technology Co., Ltd.	NA
Statistical analysis	Department of Biostatistics, Peking University Clinical Research Institute	NA
Imaging core lab	China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence	Jing Jing
	China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence	Nan Qi
CEC team	Tangdu Hospital of Air Force Medical University	Zhenwei Zhao

Collaboration team	Unit or Hospital name	Investigator Name
DSMB	Beijing Tian Tan Hospital, Capital Medical University	Kehui Dong
	The First Affiliated Hospital of Harbin Medical University	Huaizhang Shi
	People's Liberation Army Air Force General Hospital	Jin Shi
	Strategic Support Force Specialty Medical Center	Yiling Cai
AE Committee	China National Clinical Research Center for Neurological Diseases	Yuesong Pan
	Beijing Tian Tan Hospital, Capital Medical University	Xuan Sun
	Beijing Tian Tan Hospital, Capital Medical University	Yiming Deng
	Beijing Tian Tan Hospital, Capital Medical University	Ning Ma
Steering Committee	Beijing Tian Tan Hospital, Capital Medical University	Zhongrong Miao
	Beijing Tian Tan Hospital, Capital Medical University	Yilong Wang
	Beijing Tian Tan Hospital, Capital Medical University	Yongjun Wang
	Comprehensive Stroke Center, University of California, Irvine, USA	Wengui Yu
	Beijing Tian Tan Hospital, Capital Medical University	Xingquan Zhao
	Neurovascular Imaging Research Core and Department of Neurology, University of California, USA	David S Liebeskind
	Department of Neurosurgery, University Hospitals Cleveland Medical Center/ Case Western Reserve	Sepideh Amin-Hanjani
	Univeristy School of Medicine, Ohio, USA	
Cooperation team	Unit or Hospital name	Investigator Name
Sponsor	Beijing Tian Tan Hospital, Capital Medical University	NA
CRO	Beijing Jingchengtong Medicine Technology Co., Ltd.	NA
Statistical analysis	Medical Statistics Office, Peking University First Hospital, Beijing, China	NA
Imaging core lab	China National Clinical Research Center for Neurological Diseases, Tiantan Neuroimaging Center of Excellence	Jing Jing Nan Qi
CEC team	Tangdu Hospital of Air Force Medical University	Zhenwei Zhao
	Beijing Tian Tan Hospital, Capital Medical University	Kehui Dong
	The First Affiliated Hospital of Harbin Medical University	Huaizhang Shi
DSMB	People's Liberation Army Air Force General Hospital	Jin Shi
	Strategic Support Force Specialty Medical Center	Yiling Cai
	Beijing Tian Tan Hospital, Capital Medical University	Yuesong Pan
AE Committee	Beijing Tian Tan Hospital, Capital Medical University	Xuan Sun
	Beijing Tian Tan Hospital, Capital Medical University	Yiming Deng
	Beijing Tian Tan Hospital, Capital Medical University	Ning Ma

Collaboration team	Unit or Hospital name	Investigator Name
Technology	Strategic Support Force Specialty Medical Center	Yiling Cai
Steering	Xuanwu Hospital Capital Medical University	Yabing Wang
Committee	Aerospace Central Hospital	Peifu Wang
	Beijing Chaoyang Emergency Rescue Hospital of Integrated Traditional Chinese and Western Medicine	Qiang Zhang
	The Sixth Medical Center of the General Hospital of the People's Liberation Army	Feng Qiu
	Beijing Tian Tan Hospital, Capital Medical University	Zhongrong Miao
Chairs of Steering	Beijing Tian Tan Hospital, Capital Medical University	Yilong Wang
Committee	Comprehensive Stroke Center, University of California, Irvine, USA	Wengui Yu

23. Research drug supply and responsibilities division

Clopidogrel will be provided to the patients for free during the screening period of 3-5 days and the enrollment period of 90 days. Whether to continue taking clopidogrel after 90 days will be assessed according to the patient's condition, and the patient needs to buy them after 90 days.

Atorvastatin (Ale) are also free for 1 year. Whether to continue taking atorvastatin after 1 year will be assessed per the patient's condition, and patients will need to buy them after 1 year.

The uniform-free drugs used in BASIS will be stored in a safe place by researchers or other authorized personnel and stored in accordance with the conditions required (the drug instructions).

Researchers, hospital pharmacists or other persons permitted to store and distribute the research drugs are responsible for ensuring that the drugs used in clinical trials are safely maintained in accordance with the requirements of the (research) organizers and meet the corresponding regulatory requirements.

All drugs used in clinical studies should be distributed according to the researcher's prescription, and the researcher should ensure that the drugs used for research distribution and recovery are accurately recorded.

Any quality problems related to research drugs (labels, failure dates, etc.) encountered in the reception or use should be reported to the organizer in time, and the organizer should initiate a complaint procedure.

Under no circumstances will researchers provide research results to third parties, allow the research results to be used under non-clinical research protocols, or transfer the research results in any other way.

24. Research Device Processing Mechanism, Characteristics, and Responsibilities Division

Balloon devices provided by SINOMED will be used in accordance with high-quality clinical practice guidelines and medical device tracking guidelines. An introduction to the balloon device contains the following information:

Device name : Neuro RX intracranial balloon catheter

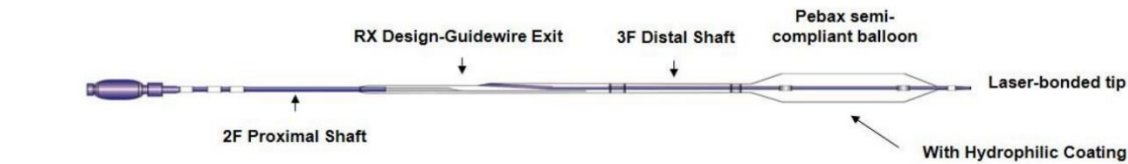
Research and development unit : Sino Medical Science and Technology Co., Ltd.

Registration number: CFDA, 20163772491 (China, Equipment, Registration, Approval, 2016-12-19)

Composition: Neuro RX intracranial balloon catheter is an expandable balloon with a rapid exchange system that can be used to treat ICAS. Its working length is 145 cm, with a guide wire of less than 0.014' and a catheter with an inner diameter of $\geq 5F$ (0.056'). Two opaque markers

on the catheter shaft indicate the working length of the balloon. The guide wire exit is 24 cm from the far tip.

Device parameters (length and diameter)



Length Diameter	6 mm	10 mm	12 mm	15 mm	20 mm	25 mm
1.50		✓		✓	✓	
2.00		✓		✓	✓	✓
2.25		✓		✓	✓	✓
2.50	✓	✓	✓	✓	✓	✓
2.75	✓	✓	✓	✓	✓	✓
3.00	✓	✓	✓	✓	✓	✓
3.25	✓	✓	✓	✓	✓	✓
3.50	✓	✓	✓	✓	✓	✓
4.00	✓	✓	✓	✓	✓	✓

Recommend: Neuro RX intracranial balloon catheter is suitable for ICAS lesion, and balloon dilatation for improving intracranial perfusion.

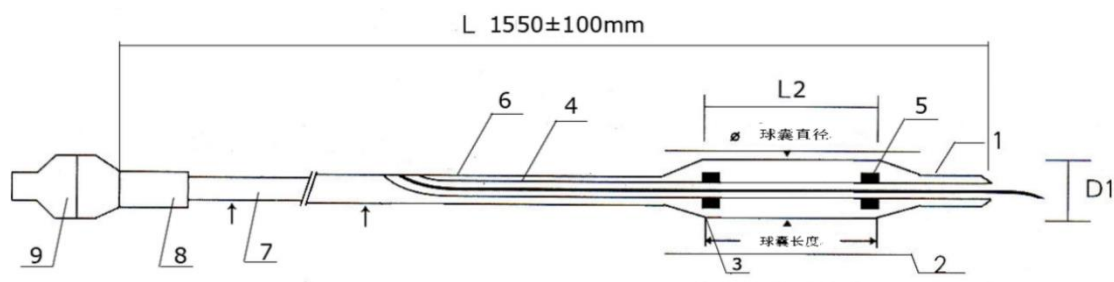
Device name: Neuro LPS intracranial balloon dilatation catheter

Research and development unit: Sino Medical Science and Technology Co., Ltd.

Registration number: NMPA, 20203030576 (China, Equipment, Registration, Approval, 2020-06-19)

Composition:

Neuro LPS intracranial balloon dilatation catheter is composed of medical polypropylene, polytetrafluoroethylene, polyether amide block copolymer, polycarbonate, polyolefin, stainless steel, and platinum alloy.



1) TIP head (polyether amide block copolymer)

2) Balloon protective sheath (polytetrafluoroethylene)

3) Balloon (109polyetherimide block copolymer)

4) Balloon inner tube (109polyetherimide block copolymer)

5) Marker (two Markers are respectively set on the balloon's inner tube (4), the outer edge of the two Markers is flush with the neck of the balloon, and the material is 90% platinum and 10% iridium alloy)

6) Balloon outer tube (109polyetherimide block copolymer)

7) Metal conduit (304 stainless steel)

8) sheath (two layers, inner layer of 304 stainless steel, outer layer of polyolefin)

9) Joint (polycarbonate)

Note 1: D1 represents the diameter of the balloon after expansion.

Note 2: L2 represents the effective length of the balloon on the balloon catheter.

Note 3: L represents the effective length of the balloon catheter.

Note 4: A complete balloon dilatation catheter is inserted into a high-density polyethylene ring protective sleeve.

Device parameters (length and diameter)

Length/Diameter	8 mm	10 mm	12 mm	15 mm	20 mm
1.50	●	●	●	●	●
2.00	●	●	●	●	●
2.25	●	●	●	●	●
2.50	●	●	●	●	●
2.75	●	●	●	●	●
3.00	●	●	●	●	●
3.25	●	●	●	●	●
3.50	●	●	●	●	●
4.00	●	●	●	●	●

Description: Neuro RX or Neuro LPS intracranial balloon dilatation catheter is provided by SINOMED, and the intracranial balloon catheter equipment is not provided free of charge.

Responsibilities Division

Investigators and/or other designated persons/institutions should archive and save the delivery records of the products to the research center, including the inventory of the research center, the use of each item, and the unused products that are returned to the sponsor. In the event of damage, loss, or additional (unscheduled) material, the organizer or relevant party shall be notified. These records should include the date, quantity, product serial number, shelf life (if any), and a unique code number assigned to the device and the research object.

25. Control requirements

BASIS will be carried out in accordance with the relevant provisions of the China Food and Drug Administration (NMPA): 'medical device clinical trial quality management norms' and 'drug clinical trial quality management norms'.

26. Institutional Review Board (IRB)/Ethics Committee (IEC)

Researchers or organizers must submit a clinical research protocol to the Ethics Committee of Beijing Tiantan Hospital and obtain a written certificate signed by the Ethics Committee with clear approval time and approval opinions.

The following information should be clearly marked on the written certificate of approval: clinical trial information (study number, name, and version number of clinical study protocol), review documents (study protocol, informed consent, etc.).

The clinical research will be initiated only after the research sponsor has received written evidence of a clear approval time and approval opinion.

During the clinical research, any revision or modification of the clinical study protocol, any event that may affect patient safety or the continuation of clinical research, especially any safety changes, and any updates to the Investigator's Manual should be reported to the Ethics Committee.

If necessary, a progress report is provided to the ethics committee each year, and the clinical research results are summarized at the end of the clinical stud

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Appendix

Appendix Table 1. ASITN/SIR collateral circulation scale

Grade 0	No collaterals visible to the ischemic site
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Grade 1	Slow collaterals to the periphery of the ischemic site with persistence of some of the defect
Grade 2	Rapid collaterals to the periphery of ischemic site with persistence of some of the defect and to only a portion of the ischemic territory
Grade 3	Collaterals with slow but complete angiographic blood flow of the ischemic bed by the late venous phase
Grade 4	Complete and rapid collateral blood flow to the vascular bed in the entire ischemic territory by retrograde perfusion

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Appendix Table 2. modified Rankin Scale

The modified Rankin Scale (mRS) is an ordinal hierarchical scale ranging from 0 to 5, with higher scores indicating more severe disability. A score of 6 has been added to signify death.

Category	Short description	Long description
0	No symptoms	No symptoms
1	Symptoms, no disability	Minor symptoms that do not interfere with lifestyle
2	Slight disability	Slight disability, symptoms that lead to some restriction in lifestyle, but do not interfere with the patient's capacity to look after himself.
3	Moderate disability	Moderate disability, symptoms that significantly restrict lifestyle and prevent totally independent existence
4	Moderately severe disability	Moderately severe disability, symptoms that clearly prevent independent existence though not needing constant attention
5	Severe disability	Severe disability, totally dependent patient requiring constant attention day and night.
6	Death	Death

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2013 **Appendix Table 3. National Institute of Health Stroke Scale (NIHSS)**

2014 The NIHSS is an ordinal hierarchical scale to evaluate the severity of stroke by assessing a
 2015 patient's performance. Scores range from 0 to 42, with higher scores indicating a more severe
 2016 deficit. Administer stroke scale items in the order listed. Record performance in each category
 2017 after each subscale exam. Do not go back and change scores. Follow directions provided for
 2018 each exam technique. Scores should reflect what the patient does, not what the clinician thinks
 2019 the patient can do. The clinician should record answers while administering the exam and work
 2020 quickly. Except where indicated, the patient should not be coached (i.e. repeated requests to
 2021 patient to make a special effort).

Instructions	Scale definition
1a. Level of consciousness. The investigator must choose a response if a full evaluation is prevented by such obstacles as an endotracheal tube, language barrier, orotracheal trauma/bandages. A 3 is scored only if the patient makes no movement (other than reflexive posturing) in response to noxious stimulation.	0 = Alert; keenly responsive. 1 = Not alert; but arousable by minor stimulation to obey, answer, or respond. 2 = Not alert; required repeated stimulation to attend, or is obtunded and requires strong or painful stimulation to make movements (not stereotyped). 3 = Responds only with reflex motor or autonomic effects or totally unresponsive, flaccid and areflexic.
1b. LOC Questions: The patient is asked the month and his/her age. The answer must be correct – there is not partial credit for being close. Phasic and stuporous patients who do not comprehend the questions will score 2. Patients unable to speak because of endotracheal intubation, orotracheal trauma, severe dysarthria from any cause, language barrier, or any other problem not secondary to aphasia are given a 1. It is important that only the initial answer be graded and that the examiners not “help” the patient with verbal or non-verbal clues.	0 = Answers both questions correctly. 1 = Answers one question correctly. 2 = Answers neither question correctly.

1c. LOC Commands: The patient is asked to open and close the eyes and then to grip and release the non-paretic hand. Substitute another one step command if the hand cannot be used. Credit is given if an unequivocal attempt is made but not completed due to weakness. If the patient does not respond to command, the task should be demonstrated to him or her (pantomime), and the result scored (i.e. follows none, one or two commands). Patients with trauma, amputation, or other physical impediments should be given suitable one-step commands. Only the first attempt is scored.	0 = Performs both tasks correctly. 1 = Performs one task correctly. 2 = Performs neither task correctly.
2. Best Gaze: Only horizontal eye movements will be tested. Voluntary or reflexive (oculocephalic) eye movements will be scored, but caloric testing is not done. If the patient has a conjugate deviation of the eyes that can be overcome by voluntary or reflexive activity, the score will be a 1. If a patient has an isolated peripheral nerve paresis (CN III, IV or VI), score a 1. Gaze is testable in all aphasic patients. Patients with ocular trauma, bandages, preexisting blindness, or other disorder of visual acuity or fields should be tested with reflexive movements, and a choice made by the investigator. Establishing eye contact and then moving about the patient from side to side will occasionally clarify the presence of a partial gaze palsy.	0= Normal. 1= Partial gaze palsy; gaze is abnormal in one or both eyes, but forced deviation or total gaze paresis is not present. 2= Forced deviation; or total gaze paresis not overcome by the oculocephalic maneuver.
3. Visual: Visual fields (upper and lower quadrants) are tested by confrontation, using finger counting or visual threat, as appropriate. Patients may be encouraged, but if they look at the side of the moving finger appropriately, this can be scored as normal. If there is unilateral blindness or enucleation, visual fields in the remaining eye are scored. Score 1 only if a clear-cut asymmetry, including quadrantanopia, is found. If patient is blind from any cause, score 3. Double simultaneous stimulation is performed in this case. If there is extinction, the patient receives a 1, and the results are used to respond to item 11.	0= No visual loss. 1= Partial hemianopia. 2= Complete hemianopia. 3= Bilateral hemianopia (blind including cortical blindness)
4. Facial palsy: Ask or use pantomime to encourage the patient to show teeth or raise eyebrows and close eyes. Score symmetry of grimace in response to noxious stimuli in the poorly response or	0 = Normal symmetrical movements. 1= Minor paralysis (flattened nasolabial fold, asymmetry on smiling) 2= Partial paralysis (total or near-total

non-comprehending patient. If facial trauma/bandages, orotracheal tube, tape or other physical barriers obscure the face, these should be removed to the extent possible.	paralysis of lower face) 3= Complete paralysis of one or both sides (absence of facial movement in the upper and lower face).
5. Motor arm: The limb is placed in the appropriate position: extend the arms (palms down) 90 degrees (if sitting) or 45 degrees (if supine). Drift is scored if the arm falls before 10 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic arm. Only in the case of amputation or joint fusion at the shoulder, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	0= No drift; limb holds 90 (or 45) degrees for full 10 seconds. 1= Drift; limb holds 90 (or 45) degrees, but drifts down before full 10 seconds; does not hit bed or other support. 2= Some effort against gravity; limb cannot get to or maintain (if cued) 90 (or 45) degrees, drifts down to bed, but has some effort against gravity. 3= No effort against gravity; limb falls. 4= No movement. UN = Amputation or joint fusion: explain: 5a = Left Arm. 5b = Right arm.
6. Motor leg: The limb is placed in the appropriate position: hold the leg at 30 degrees (always tested supine). Drift is scored if the leg falls before 5 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic leg. Only in the case of amputation or joint fusion at the hip, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	0= No drift; leg holds 30-degree position for full 5 seconds. 1= Drift; leg falls by the end of the 5-second period but does not hit bed. 2= Some effort against gravity; leg falls to bed by 5 seconds, but has some effort against gravity. 3= No effort against gravity; leg falls to bed immediately. 4= No movement. UN = Amputation or joint fusion: explain: 6a. Left Leg 6b. Right Leg.
7. Limb ataxia: This item is aimed at finding evidence of a unilateral cerebellar lesion. Test with eyes open. In case of visual defect, ensure testing is done in intact visual field. The finger-nose-finger and heel-shin tests are performed on both sides, and ataxia is scored only if present out of proportion to weakness. Ataxia is absent in the patient who cannot understand or is paralyzed. Only in the case of amputation or joint fusion, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice. In case of blindness, test by having the	0= Absent. 1= Present in one limb. 2= Present in two limbs. UN = Amputation or joint fusion: explain:

patient touch nose from extended arm position.	
8. Sensory: Sensation or grimace to pinprick when tested, or withdrawal from noxious stimulus in the obtunded or aphasic patient. Only sensory loss attributed to stroke is scored as abnormal and the examiner should test as many body areas (arms [not hands], legs, trunk, face) as needed to accurately check for hemisensory loss. A score of 2, 'severe or total sensory loss', should only be given when a severe or total loss of sensation can be clearly demonstrated. Stuporous and aphasic patients will, therefore, probably score 1 or 0. The patient with brainstem stroke who has bilateral loss of sensation is scored 2. If the patient does not respond and is quadriplegic, score 2. Patients in a coma (item 1a=3) are automatically given a 2 on this item.	0= Normal; no sensory loss. 1= Mild-to-moderate sensory loss; patients feels pinprick is less sharp or is dull on the affected side; or there is a loss of superficial pain with pinprick, but patient is aware of being touched. 2= Severe to total sensory loss; patient is not aware of being touched in the face, arm and leg.
9. Best language: A great deal of information about comprehension will be obtained during the preceding sections of the examination. For this scale item, the patient is asked to describe what is happening in the attached picture, to name the items on the attached naming sheet and to read from the attached list of sentences. Comprehension is judged from responses here, as well as to all of the commands in the preceding general neurological exam. If visual loss interferes with the tests, ask the patient to identify objects placed in the hand, repeat, and produce speech. The intubated patient should be asked to write. The patient in a coma (item 1a=3) will automatically score 3 on this item. The examiner must choose a score for the patient with stupor or limited cooperation, but a score of 3 should be used only if the patient is mute and follows no one-step commands.	0= No aphasia; normal 1= Mild-to-moderate aphasia; some obvious loss of fluency or facility of comprehension, without significant limitation on ideas expressed or form of expression. Reduction of speech and/or comprehension, however, makes conversation about provided materials difficult or impossible. For example, in conversation about provided materials, examiner can identify picture or naming card content from patient's response. 2= Severe aphasia; all communication is through fragmentary expression; great need for inference, questioning, and guessing by the listener. Range of information that can be exchanged is limited; listener carries burden of communication. Examiner cannot identify materials provided from patient response. 3 = Mute, global aphasia: no usable speech or auditory comprehension.
10. Dysarthria: If patient is thought to be normal, an adequate sample of speech must be obtained by asking patient to read or repeat words from the attached list.	0= Normal. 1= Mild-to-moderate dysarthria; patient slurs at least some words and, at worst,

If the patient has severe aphasia, the clarity of articulation of spontaneous speech can be rated. Only if patient is intubated or has other physical barriers to producing speech, the examiner should record the score as untestable (UN), and clearly write an explanation for this choice. Do not tell the patient why he or she is being tested.	can be understood by some difficulty. 2= Severe dysarthria: patient's speech is so slurred as to be unintelligible in the absence of or out of proportion to any dysphasia, or is mute/anarthric. UN = Intubated or other physical barrier.
11. Extinction and Inattention (formerly Neglect): Sufficient information to identify neglect may be obtained during the prior testing. If the patient has a severe visual loss preventing visual double simultaneous stimulation, and the cutaneous stimuli are normal, the score is normal. If the patient has aphasia but does appear to attend to both sides, the score is normal. The presence of visual spatial neglect or anosagnosia may also be taken as evidence of abnormality. Since the abnormality is scored only if present, the item is never untestable.	0= No abnormality. 1= Visual, tactile, auditory, spatial, or personal inattention or extinction to bilateral simultaneous stimulation in one of the sensory modalities. 2= Profound hemi-inattention or extinction to more than one modality; does not recognize own hand or orients to only one side of space.

2023 **Appendix Table 4. EuroQoL 5D-5L**

2024 Under each heading, please tick the ONE box that best describes your health TODAY.

2025 **Mobility**

I have no problems in walking about ☐

I have slight problems in walking about ☐

I have moderate problems in walking about ☐

I have severe problems in walking about ☐

I am unable to walk about ☐

Self-care

I have no problems washing or dressing myself ☐

I have slight problems washing or dressing myself ☐

I have moderate problems washing or dressing myself ☐

I have severe problems washing or dressing myself ☐

I am unable to wash or dress myself ☐

Usual activities (e.g. work, study, housework, family or leisure activities)

I have no problems doing my usual activities ☐

I have slight problems doing my usual activities ☐

I have moderate problems doing my usual activities ☐

I have severe problems doing my usual activities ☐

I am unable to do my usual activities ☐

Pain/discomfort

I have no pain or discomfort ☐

I have slight pain or discomfort ☐

I have moderate pain or discomfort ☐

I have severe pain or discomfort ☐

I have extreme pain or discomfort ☐

Anxiety/depression

I am not anxious or depressed ☐

I am slightly anxious or depressed ☐

I am moderately anxious or depressed ☐

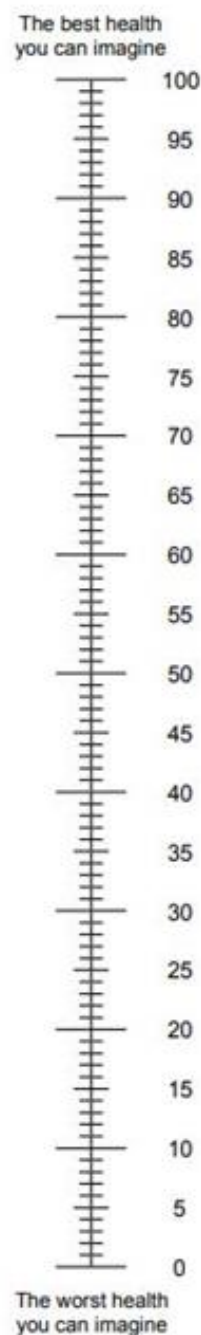
I am severely anxious or depressed ☐

I am extremely anxious or depressed ☐

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- We would like to know how good or bad your health is TODAY.
- This scale is numbered from 0 to 100.
- 100 means the best health you can imagine.
- 0 means the worst health you can imagine.
- Mark an X on the scale to indicate how your health is TODAY.
- Now, please write the number you marked on the scale in the box below.

YOUR HEALTH TODAY =



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Appendix Table 5. mTICI grade

mTICI grade	Definitions
0 grade	No perfusion
1 grade	Antegrade reperfusion past the initial occlusion, but limited distal branch filling with little or slow distal reperfusion
2a grade	Antegrade reperfusion of less than half of the occluded target artery previously ischemic territory (eg, in 1 major division of the MCA and its territory)
2b grade	Antegrade reperfusion of more than half of the previously occluded target artery ischemic territory (eg, in 2 major divisions of the MCA and their territories)
3 grade	Complete antegrade reperfusion of the previously occluded target artery ischemic territory, with absence of visualized occlusion in all distal branches

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Summary of changes

2044 BASIS Protocol V2.0-20180625 (original version)

2045 BASIS Protocol V2.1-20200414 (amended version)

2046 BASIS Protocol V2.2-20211222 (amended version)

2047 BASIS Protocol V2.3-20220328 (final version)

Version	Date of revised	Changed	Reasons for change
V2.1	2020/04/14	Change the interval between stroke onset to enroll in inclusion criteria (1, 6 and 8.1) Participants with primary or recurrent sICAS (a recent TIA [<90 days] or ischemic stroke [14-90 days] before enrollment attributed to 70-99% atherosclerotic stenosis of a major intracranial artery)	According to the Chinese Expert Consensus 2018 on Endovascular Treatment of Intracranial Atherosclerotic Stenosis, (ICAS) recommended that 2 weeks endovascular treatment after acute ischemic stroke may be safe.
		Clarify the inclusion criteria of lesion artery (1 and 8.1): 3. Diagnosed by DSA: a major intracranial artery (terminal internal carotid artery [ICA] [C4-C7 segments], middle cerebral artery [MCA] M1 segment, vertebral artery [VA] V4 segment, and basilar artery [BA])	Clear the specific lesion location of anterior and posterior circulation
V2.2	2021/12/22	Clarify exclusion criteria (1 and 8.2) 6. Target artery's supplying artery stenosis $> 50\%$. For example, patient with MCA severe stenosis (target artery) and the ipsilateral ICA stenosis $> 50\%$ should be excluded; patient with BA severe stenosis (target artery) and the dominant VA stenosis $> 50\%$	Clarify exclusion criteria

		should be excluded; the patient with extracranial artery (non-lesion side) stenosis > 70% should be excluded; For patients with balanced VA, if bilateral VA stenosis > 70% at the same time, should be excluded (unable to determine which VA is the lesion artery). However, if patient's dominant VA is the lesion artery with a dysplasia or slender contralateral VA or a non-dominant contralateral VA terminating at posteroinferior cerebellar artery, should not be excluded.	
		Add 24-month and 36-month follow-up outcomes to secondary outcomes (1 and 7.2) 10. Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target artery or all-cause deaths within 24 months after enrollment; 11. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 24 months after enrollment; 12. mRS at 24 months; 13. Combined events such as stroke, myocardial infarction, and vascular death within 24 months after enrollment; 14. Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target artery or all-cause deaths within 36 months after enrollment; 15. Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery within 36 months after enrollment; 16. Neurological improvement assessed by mRS at 36 months; 17. Combined events such as stroke, myocardial infarction, and vascular death within 36 months after enrollment.	To explore the longer-term effect of BA for sICAS.

		Re-estimate the sample size (1 and 20.1) Change the sample size from 802 to 512 (Details in SAP 2.0)	According to more recent studies and the investigator's clinical practice in China, the composite event rate in the AMM group was 15% and 7% in the BA group. With the updated data of events rates and considering the difficulty in enrolling patients related to the COVID-19 pandemic and insufficient funding, the data safety and monitoring board recommended to forgo the interim analysis
V2.3	2022/03/28	Add new outcomes to secondary outcomes (1 and 7.2) 1. Any stroke (ischemic or hemorrhage stroke) or all-cause deaths within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up. 6.Revascularization of the target artery within 12 months after enrollment;	The two outcomes were also important for the BASIS trial.

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**Balloon Angioplasty vs. Aggressive Medical Management for
Symptomatic Intracranial Artery Stenosis: The BASIS Randomized
Clinical Trial
(BASIS)**



Statistical Analysis Plan

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Version 1.0

Date: December 27, 2018

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1. Introduction

This statistical analysis plan (SAP) documents the planned statistical analyses for the BASIS study and is based on the protocol, together with any subsequent amendments. This SAP is intended for the use of project team members and should be read in conjunction with the aforementioned protocol.

2. Study Objective

To investigate whether balloon angioplasty combined with aggressive medical management (AMM) is superior to AMM alone for patients with high-risk intracranial atherosclerotic stenosis (sICAS) in the primary endpoint (ischemic/hemorrhage stroke or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization from the qualifying artery beyond 30 days through 12 months after enrollment).

3. Study Endpoint(s)

3.1. Primary Endpoint

The primary endpoint is stroke or death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization of the qualifying artery beyond 30 days through 12 months after enrollment.

3.2. Secondary Endpoints

- ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target artery or all-cause deaths within 90 days after enrollment;

- 2145 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery
2146 within 90 days after enrollment;
- 2147 ✓ mRS (modified Rankin Scale) at 90 days;
- 2148 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target artery or
2149 all-cause deaths within 12 months after enrollment;
- 2150 ✓ Revascularization of the qualifying artery within 1 year after enrollment;
- 2151 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery
2152 within 12 months after enrollment;
- 2153 ✓ mRS at 12 months;
- 2154 ✓ Restenosis rate of the target artery within 12 months (defined as rate of stenosis >70% or
2155 increased by 30% based on following-up neurovascular imaging);
- 2156 ✓ Combined events such as stroke, myocardial infarction, and vascular death within 12
2157 months after enrollment;
- 2158 ✓ Life quality assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire) within 12
2159 months after enrollment;

2160 3.3. Safety Assessment

2161 From enrollment to 1 year, all patients were evaluated for safety.

- 2162 ✓ Vital signs: blood pressure, heart rate, respiration rate, body temperature.
- 2163 ✓ Blood laboratory test includes blood routine (red blood cells, white blood cells, platelets,
2164 hemoglobin), biochemistry examinations (alanine aminotransferase, aspartate
2165 aminotransferase, alkaline phosphatase, glutamylphthalase, total bilirubin, urea nitrogen,

2166 serum creatinine), coagulation function, hepatic and renal function tests, fasting blood
2167 glucose and HbA1c.

2168 ✓ Adverse events and serious adverse events.

2169

2170 **4. Study Hypothesis**

2171 Compared to AMM alone, balloon angioplasty combined with AMM could decrease 50% risk of
2172 the primary endpoint in patients with high-risk sICAS during over 12-month follow-up period.

2173

2174 **5. Design**

2175 **5.1 Study Design**

2176 After the study is formally started, the participating study centers copied the video (including
2177 the DSA and surgical process) of 3-5 cases of balloon angioplasty in their centers and transfer
2178 the video to the core laboratory (Beijing Tiantan Hospital). After the consistency is passed, the
2179 group continued to be enrolled, so as to ensure the consistency of the process of balloon
2180 angioplasty.

2181 BASIS trial is an investigator-initiated, multicenter, prospective, randomized, open-label,
2182 blinded end-point (PROBE) trial that plans to enroll the patient with sICAS (defined as a recent
2183 TIA (<90 days) or ischemic stroke (21-90 days) attributed to a 70-99% atherosclerotic stenosis
2184 of a major intracranial artery (anterior/posterior circulation). The patients are randomly assigned
2185 to either the experimental group or the control group at 1:1 ratio.

2186 Experimental group: patients will receive balloon angioplasty (details in procedure part) plus
2187 AMM (same as the control group);

2188 Control group: patients will receive the AMM alone.

2189 ✓ Drug administration plan of AMM: aspirin 100 mg daily for the entire follow-up and
2190 clopidogrel 75mg daily for the first 90 days after enrollment. Whether to continue taking
2191 clopidogrel after 90 days per the condition of the patient, clopidogrel can be replaced with
2192 ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance;
2193 atorvastatin calcium tablets were used 20mg-80mg/day for one year after enrollment; the
2194 dose was adjusted to control LDL <1.8mmol/L or <70mg/dl. Whether to continue taking
2195 atorvastatin calcium tablets after one year per the patient's condition.

2196 ✓ Risk factors management—Control blood pressure 130-140/90-100 mmHg and LDL
2197 <1.8mmol/L or <70mg/dl

2198 The first 100 participants completed the peri-operative (30 days) observation, mainly collecting
2199 the adverse events during the peri-operative period. PI needs to organize other investigators
2200 from the participating centers for interim discussion to evaluate the feasibility of this treatment
2201 plan and the continuation of later research, and follow-up research can be conducted later.

2202 5.2 Planned Analyses

2203 The analyses that are detailed in this SAP will be performed only when the database has been
2204 locked, all protocol violators identified, and treatment allocations have been unblinded.
2205 Membership of the Full Analysis and Per Protocol populations will be determined using the
2206 rules set out in this SAP. At a date to be agreed within the project team, a data look will be

performed. This will involve production of all data displays on a subset of the data using dummy treatment codes. These are produced purely as an aide to the pre-programming of the study and no unblinding will occur.

5.3 Interim Analyses

Interim analysis is performed when half of patients finished 12-months visit. The mid-term analysis α_1 is estimated to be 0.0015, and the final analysis α_2 is 0.024.

6. Sample Size Estimates

The estimation of the sample size is calculated according to the 1-year results of the SAMMPRIS trial, the incidence of target vessel stroke or death rate was 12.2% within 12 months after enrollment in the experimental group. The sample size was 686 cases, with a 80% power, two-sided $\alpha = 0.05$, to detect a 50 % relative reduction (6.1%) after balloon angioplasty for intracranial atherosclerosis. It is assumed that the O'Brien-Fleming spending function will lead to 6% expansion of the total type I error. The SAS software is used to calculate the sample size required for the mid-term analysis, which increases the sample size to the required 728 cases. Considering the 10% lost to follow-up rate, the final sample size of this study is 802 cases.

7. Analysis Populations

7.1 Intention-to-treat Set (ITT)

All patients who have been randomized to groups and receive the study intervention will be included. And the analyses will be performed based on the intervention as per the patient randomized result, according to the basic principle of intention-to-treat (ITT).

2228 This population will be the primary population for analyses of efficacy.

2229 7.2. Per Protocol Set (PPS)

2230 Per Protocol Set (PPS) is a subset of FAS (mITT). All patients with finishing the treatment or
2231 without violating the trail program seriously are included in PPS. The exact definition of a
2232 serious violation will be finalized at the time of data review and may generally include (but is not
2233 limited to) the following criteria: complied with the experimental protocol, had good compliance,
2234 did not use prohibited drugs during the experiment, and completed the contents specified in the
2235 case report form. A partial protocol violator will be included in the Per Protocol Population up to
2236 the time of their violation. Serious violations will be determined during data review. PPS is a
2237 secondary data set for the end point evaluation in this study.

2238 7.3. As-treated Set (ATS)

2239 All patients who have been randomized to groups with more than one use of study products will
2240 be included. And the analyses will be performed based on the intervention as per the patient
2241 received. ATS is a secondary data set for the end point evaluation in this study.

2242 7.4. Safety Set (SS)

2243 All patients who received at least 1-time of study products and safety assessment available will
2244 be included in the safety population. This population will be used for safety analyses.

2245

2246 8. Treatment Comparisons

2247 The treatment comparison of interest in this study is to assess the effect on reducing the risk of

primary endpoint (ischemic/hemorrhage stroke or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization from the qualifying artery beyond 30 days through 12 months after enrollment) of balloon angioplasty combined with AMM versus AMM alone patients with high-risk sICAS.

9. General Considerations for Data Analyses

All programming will be performed using SAS Version 9.4. All analysis output will use the following treatment group naming conventions and treatment order:

Experimental group: patients will receive balloon angioplasty (details in procedure part) plus AMM (same as the control group);

Control group: patients will receive the AMM alone.

All statistics were two sided with a $P < 0.05$ considered significant.

9.1 Multicenter Studies

The study will include a large number of centers to participate in this trial, some centers will have limited subjects per center in this trial, in main analyses won't consider center-effects adjustment. The analyses for consider center-effects adjustment will serve as sensitivity analyses. Centers with small sample size of subjects will be pooled with larger centers so that centers are of a reasonable size for the purpose of the statistical analyses. Cox proportional hazards model will be adopted to calculate HR for the treatment effect considering center-effects adjustment.

9.2 Examination of Subgroups

The rate of primary outcome will be presented for each level of the covariates listed below. The extent to which the treatment effect varies across levels of each subgroup will be assessed through interaction tests.

9.3 Multiple Comparisons and Multiplicity

A single primary efficacy variable has been defined for this study, with all other efficacy variables identified as secondary or other. Similarly, only one treatment comparison is of interest in the study and therefore there are no requirements to adjust for multiple comparisons or multiple endpoints within this study.

10.Data Handling Conventions

10.1 Premature Withdrawal and Missing Data

If any subject withdraws prematurely from the study (prior to the final visit), they are required to complete the withdrawal visit in the CRF. The reasons for withdrawal will be presented in a summary table. For the purposes of summaries and analysis of clinic visit data, this visit will be assigned to the next scheduled clinic visit for that subject, regardless of whether the date falls within the next visit window.

Subjects who withdraw before the end of the study, but who do provide at least one post-baseline measure for a particular endpoint, will be included in the analysis.

Subjects who do not attend any visits after randomization will be excluded from analysis of any endpoint, as no post-baseline data will be available.

10.2 Event Rates

The number of people and person-time of events should be recorded in detail and showing the event rate of each treatment group in summary statement.

The event rate for each treatment group will be calculated as: the sum of number of event for all the patients / the sum of number of treatment periods for all the patients.

10.3 Time to Event Analysis

Differences between treatments in the risk of any stroke (ischemic or haemorrhagic) event or

combined vascular events during follow-up were assessed using standard Kaplan-Meier time-to-event approaches. The time to the first event was used in the model when there were multiple events of the same type. Patients were considered censored at the time of study termination or death if there were no events occurred during the study.

11.Study Population

11.1 Disposition of Subjects

The number of subjects in each analysis population will be presented, subjects to be excluded from the Per Protocol population will be listed, and the total number of subjects attending each clinic visit will also be summarized by treatment group.

The number of subjects randomized, completed and prematurely withdrawn from the study will be presented for each treatment group. The primary reasons for withdrawal both prior to and post randomization will also be presented.

A data display listing and summary of deviations from the inclusion/exclusion criteria will be presented for all subjects who were either entered or randomized into the trial.

11.2 Protocol Deviations

Subject data will be examined for evidence of protocol violators in order to assess how well the protocol was followed. Inclusion and exclusion criteria are detailed in the study protocol.

Subjects who commit protocol violations will be included in the FAS (mITT) Population but excluded from the Per Protocol Population. These protocol violations will be shown in a listing. Subjects can either be full or partial protocol violators. A full protocol violator is completely excluded from the Per Protocol Population. A partial protocol violator has only some data excluded. For subjects who violated the protocol during the treatment period due to unpermitted changes in the medication or prohibited concurrent medication, the analysis will only use data recorded prior to the violation. For all violations which reference the treatment period, the treatment start date will be used as the reference date.

A listing of all possible protocol violators will be produced for clinical review. The final list of subjects who are protocol violators and are therefore excluded from the Per-Protocol population will be agreed by the study team.

12. Statistical Analysis

12.1 Demographic and Baseline Characteristics

The following demographic information will be listed and summarized for subjects in each treatment group: age, sex, height, weight, vital signs. This will also be done for medical history, medication history, stroke and TIA history.

The continuous data will be presented as median and interquartile range; categorical data will be presented as n(%). Wilcoxon rank sum test will be used for comparison between two continuous data, and Chi-squared tests or Fisher exact test will be used for comparison between two categorical data.

12.2 Primary Efficacy Endpoint

The composite event rates and corresponding 95% CIs of the primary outcome in the two treatment groups will be estimated by Kaplan-Meier survival analysis and compared by log-rank test. Hazard ratio between the two groups and its 95% CI will be calculated using a Cox proportional hazards regression model.

The main analysis of this study will be conducted based on intention-to-treat principle, and a per protocol analysis and as-treated analysis will also be conducted as sensitivity analysis. ITT, PPS and ATS will be used to analyze for the primary efficacy outcome, respectively. If the results in the PPS and ATS population are inconsistent with the ITT population, detailed analysis

2344 of the inconsistent results is required.

2345 **Subgroup Analysis**

2346 Summary tables will be produced for the predefined subgroups and interactions between
2347 treatment and these subgroups will be investigated, using a Cox proportional hazards model. A
2348 separate model will be used for each interaction to determine its significance. This will also be
2349 presented graphically on a forest plot. Treatment effect modification on the primary endpoint
2350 will be explored in the following subgroups:

- 2351 ✓ Age, <65 years old vs. ≥ 65 years old
- 2352 ✓ Sex, men vs. women
- 2353 ✓ Hypertension, yes vs. no
- 2354 ✓ Diabetes, yes vs. no
- 2355 ✓ Smoking, yes vs. no
- 2356 ✓ Baseline renal function, $\text{eGFR} < 60 \text{ ml/min/1.73m}^2$, $\text{eGFR} \geq 60 \text{ ml/min/1.73m}^2$
- 2357 ✓ Target vessel stenosis degree, $< 80\%$, $\geq 80\%$
- 2358 ✓ BMI, $< 25 \text{ Kg/m}^2$, $25\text{-}30 \text{ Kg/m}^2$, $\geq 30 \text{ Kg/m}^2$
- 2359 ✓ Hypoperfusion, CTP, ASITN/SIR collateral circulation scale
- 2360 ✓ Anterior circulation vs. posterior circulation
- 2361 ✓ The mechanism, ischemic stroke vs. TIA

2362

2363 **12.3 Secondary Efficacy Endpoint**

- 2364 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target artery or
2365 all-cause deaths (within 90 days and 12 months after enrollment) after enrollment will be

2366 analyzed using a Cox proportional hazard model. The hazard ratios with 95% CIs will be
2367 reported;

2368 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target artery
2369 (within 90 days and 12 months after enrollment) will be analyzed using a Cox
2370 proportional hazard model. The hazard ratios with 95% CIs will be reported;

2371 ✓ Common odds ratios of mRS (at 90 days and 12 months) will be estimated using ordinal
2372 logistic regression.

2373 ✓ Logistic regression analysis will be performed to compare the restenosis rate of the target
2374 artery within 12 months (defined as rate of stenosis >70% or increased by 30% based on
2375 following-up neurovascular imaging) between the two treatment groups and the odds
2376 ratio with 95% CI will be reported.

2377 ✓ Combined events such as stroke, myocardial infarction, and vascular death within 12
2378 months after enrollment will be analyzed using a Cox proportional hazard model. The
2379 hazard ratios with 95% CIs will be reported.

2380 ✓ Life quality assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire) within 12
2381 months after enrollment will be tested using student t-test or Wilcoxon rank sum test as
2382 appropriate.

2383 12.4 Safety Analyses

2384 Adverse Events

2385 Adverse events (AEs) will be coded using the MedDRA coding dictionary (Version 26 or a later
2386 release) and grouped by system organ class. Separate data display listings and summaries will be

presented for adverse events. Within each treatment group, the number and percentage of subjects experiencing an AE will be summarized by system organ class and preferred term and Fisher's Exact test will be used to compare the number of each grouped AE event between treatment groups.

Deaths and Serious Adverse Events

Summary tables and data displays will be provided for serious adverse events (as detailed in the study protocol). In addition, all deaths and serious AE's will be documented in a case narrative format in the clinical study report.

The number of deaths occurring over the treatment period will be summarized and Fisher's Exact test will be used to compare the number of deaths between treatment groups.

Vital Signs

The results of vital signs at different time points will be statistically described and compared.

12.5 Concomitant Medication

Describe the list of concomitant medication including antiplatelet drug (aspirin, clopidogrel, ticagrelor, cilostazol), anticoagulation drug, lipid-lowering drug (statin), antihypertensive drugs (diuretic, angiotensin-converting enzyme inhibitor, angiotensin receptor blocker, beta-blocker, calcium channel antagonist, central α agonist, vasodilator) and so on. The Chi-squared tests or Fisher exact test will be used to compare the number between treatment groups.

13.Reference

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**Balloon Angioplasty vs. Aggressive Medical Management for
Symptomatic Intracranial Artery Stenosis: The BASIS Randomized
Clinical Trial
(BASIS)**



Statistical Analysis Plan

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1. Introduction

This statistical analysis plan (SAP) documents the planned statistical analyses for the BASIS study and is based on the protocol, together with any subsequent amendments. This SAP is intended for the use of project team members and should be read in conjunction with the aforementioned protocol.

2. Study Objective

To investigate whether balloon angioplasty combined with aggressive medical management (AMM) is superior to AMM alone for patients with high-risk intracranial atherosclerotic stenosis (sICAS) in the primary endpoint (ischemic/hemorrhage stroke or all-cause death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization from the qualifying artery beyond 30 days through 12 months after enrollment).

3. Study Endpoint(s)

3.1. Primary Endpoint

The primary endpoint is stroke or death within 30 days after enrollment or after balloon angioplasty procedure of the qualifying lesion during follow-up, or any ischemic stroke or revascularization of the qualifying artery beyond 30 days through 12 months after enrollment.

3.2. Secondary Endpoints

✓ Any stroke (ischemic or hemorrhage stroke) or all-cause deaths within 30 days after

- 2515 enrollment or after balloon angioplasty procedure of the qualifying lesion during
2516 follow-up
- 2517 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target
2518 artery or all-cause deaths within 90 days after enrollment;
- 2519 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target
2520 artery within 90 days after enrollment;
- 2521 ✓ mRS at 90 days;
- 2522 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target
2523 artery or all-cause deaths within 12 months after enrollment;
- 2524 ✓ Revascularization of the target artery within 12 months after enrollment;
- 2525 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target
2526 artery within 12 months after enrollment;
- 2527 ✓ mRS at 12 months;
- 2528 ✓ Restenosis rate of the target artery within 12 months (defined as rate of stenosis >70%
2529 or increased by 30% based on following-up neurovascular imaging);
- 2530 ✓ Combined events such as stroke, myocardial infarction, and vascular death within
2531 12 months after enrollment;
- 2532 ✓ Life quality assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire)
2533 within 12 months after enrollment;
- 2534 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target
2535 artery or all-cause deaths within 24 months after enrollment;
- 2536 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target

- 2537 artery within 24 months after enrollment;
- 2538 ✓ mRS at 24 months;
- 2539 ✓ Combined events such as stroke, myocardial infarction, and vascular death within
- 2540 24 months after enrollment;
- 2541 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target
- 2542 artery or all-cause deaths within 36 months after enrollment;
- 2543 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target
- 2544 artery within 36 months after enrollment;
- 2545 ✓ Neurological improvement assessed by mRS at 36 months;
- 2546 ✓ Combined events such as stroke, myocardial infarction, and vascular death within
- 2547 36 months after enrollment.

2548 3.3. Safety Assessment

- 2549 From enrollment to 1 year, all patients were evaluated for safety.
- 2550 ✓ Vital signs: blood pressure, heart rate, respiration rate, body temperature.
- 2551 ✓ Blood laboratory test includes blood routine (red blood cells, white blood cells,
- 2552 platelets, hemoglobin), biochemistry examinations (alanine aminotransferase,
- 2553 aspartate aminotransferase, alkaline phosphatase, glutamylphthalase, total bilirubin,
- 2554 urea nitrogen, serum creatinine), coagulation function, hepatic and renal function
- 2555 tests, fasting blood glucose and HbA1c.
- 2556 ✓ Adverse events and serious adverse events.

2557

2558 4. Study Hypothesis

Compared to AMM alone, balloon angioplasty combined with AMM could decrease 50% risk of the primary endpoint in patients with high-risk sICAS during over 12-month follow-up period.

5. Design

5.1 Study Design

After the study is formally started, the participating study centers copied the video (including the DSA and surgical process) of 3-5 cases of balloon angioplasty in their centers and transfer the video to the core laboratory (Beijing Tiantan Hospital). After the consistency is passed, the group continued to be enrolled, so as to ensure the consistency of the process of balloon angioplasty.

BASIS trial is an investigator-initiated, multicenter, prospective, randomized, open-label, blinded end-point (PROBE) trial that plans to enroll the patient with sICAS (defined as a recent TIA or ischemic stroke attributed to a 70-99% atherosclerotic stenosis of a major intracranial artery including terminated internal carotid artery [ICA] C4-C7 segment, middle cerebral artery [MCA] M1 segment, vertebral artery [VA] V4 segment and basilar artery [BA]) determined by DSA. The patients are randomly assigned to either the experimental group or the control group at 1:1 ratio.

Experimental group: patients will receive balloon angioplasty (details in procedure part) plus AMM (same as the control group);

Control group: patients will receive the AMM alone.

✓ Drug administration plan of AMM: aspirin 100 mg daily for the entire follow-up

and clopidogrel 75mg daily for the first 90 days after enrollment. Whether to continue taking clopidogrel after 90 days per the condition of the patient, clopidogrel can be replaced with ticagrelor or cilostazol with appropriate dosage for patients with clopidogrel resistance; atorvastatin calcium tablets were used 20mg-80mg/day for one year after enrollment; the dose was adjusted to control LDL <1.8mmol/L or <70mg/dl. Whether to continue taking atorvastatin calcium tablets after one year per the patient's condition.

- ✓ Risk factors management—Control blood pressure 130-140/90-100 mmHg and LDL <1.8mmol/L or <70mg/dl

The first 100 participants completed the peri-operative (30 days) observation, mainly collecting the adverse events during the peri-operative period. PI needs to organize other investigators from the participating centers for interim discussion to evaluate the feasibility of this treatment plan and the continuation of later research, and follow-up research can be conducted later.

5.2 Planned Analyses

The analyses that are detailed in this SAP will be performed only when the database has been locked, all protocol violators identified, and treatment allocations have been unblinded. Membership of the Full Analysis and Per Protocol populations will be determined using the rules set out in this SAP. At a date to be agreed within the project team, a data look will be performed. This will involve production of all data displays on a subset of the data using dummy treatment codes. These are produced purely as an aide to the pre-programming of the study and no unblinding will occur.

5.3 Interim Analyses

Interim analyses will not be performed in this trial. However, a Data and Safety Monitoring Board (DSMB) is in place to ensure the safety of subjects in the study. An independent statistician will prepare summary statistics on enrollment, subject status in the study, baseline characteristics, and safety data, including summary tables of coded SAEs and these will be examined by the DSMB. These tables will be provided to the DSMB at regular intervals. If the tables give rise to safety concerns for any treatment, the DSMB may recommend that the trial should be modified or stopped prematurely. The Steering Committee will, in conjunction with the Sponsor, decide whether to act on this recommendation.

6. Sample Size Estimates

According to VISSIT trial and a previous randomized trial in China, the composite event rate of the primary outcome in the medical group is anticipated to be 15%. As to the balloon angioplasty group, we assume a 7% of the primary outcome based on studies of angioplasty without stenting and investigators' clinical practice experience in China. As a result, the sample size required to detect an absolute difference of 8%. A total of 512 patients (256 per group) will be enrolled considering an 80% statistical power at one-sided α of 2.5% and a 10% dropout rate.

7. Analysis Populations

7.1 Intention-to-treat Set (ITT)

All patients who have been randomized to groups and receive the study intervention will

be included. And the analyses will be performed based on the intervention as per the patient randomized result, according to the basic principle of intention-to-treat (ITT).

This population will be the primary population for analyses of efficacy.

7.2. Per Protocol Set (PPS)

Per Protocol Set (PPS) is a subset of FAS (mITT). All patients with finishing the treatment or without violating the trial program seriously are included in PPS. The exact definition of a serious violation will be finalized at the time of data review and may generally include (but is not limited to) the following criteria: complied with the experimental protocol, had good compliance, did not use prohibited drugs during the experiment, and completed the contents specified in the case report form. A partial protocol violator will be included in the Per Protocol Population up to the time of their violation. Serious violations will be determined during data review. PPS is a secondary data set for the end point evaluation in this study.

7.3. As-treated Set (ATS)

All patients who have been randomized to groups with more than one use of study products will be included. And the analyses will be performed based on the intervention as per the patient received. ATS is a secondary data set for the end point evaluation in this study.

7.4. Safety Set (SS)

All patients who received at least 1-time of study products and safety assessment available will be included in the safety population. This population will be used for safety

2647 analyses.

2648

2649 **8. Treatment Comparisons**

2650 The treatment comparison of interest in this study is to assess the effect on reducing the
2651 risk of primary endpoint (ischemic/hemorrhage stroke or all-cause death within 30 days
2652 after enrollment or after balloon angioplasty procedure of the qualifying lesion during
2653 follow-up, or any ischemic stroke or revascularization from the qualifying artery beyond
2654 30 days through 12 months after enrollment) of balloon angioplasty combined with
2655 AMM versus AMM alone patients with high-risk sICAS.

2656

2657 **9. General Considerations for Data Analyses**

2658 All programming will be performed using SAS Version 9.4. All analysis output will use
2659 the following treatment group naming conventions and treatment order:

2660 Experimental group: patients will receive balloon angioplasty (details in procedure part)
2661 plus AMM (same as the control group);

2662 Control group: patients will receive the AMM alone.

2663 All statistics were two sided with a $P < 0.05$ considered significant.

2664 **9.1 Multicenter Studies**

2665 The study will include a large number of centers to participate in this trial, some centers
2666 will have limited subjects per center in this trial, in main analyses won't consider

center-effects adjustment. The analyses for consider center-effects adjustment will serve as sensitivity analyses. Centers with small sample size of subjects will be pooled with larger centers so that centers are of a reasonable size for the purpose of the statistical analyses.

In multicenter randomized controlled clinical study, there were some difference effect in different center due to different baseline, clinical practice or other factor, therefore, center effect analysis was required. Cox proportional hazards model will be adopted to calculate HR for the treatment effect considering center-effects adjustment.

9.2 Examination of Subgroups

The rate of primary outcome will be presented for each level of the covariates listed below. The extent to which the treatment effect varies across levels of each subgroup will be assessed through interaction tests.

9.3 Multiple Comparisons and Multiplicity

A single primary efficacy variable has been defined for this study, with all other efficacy variables identified as secondary or other. Similarly, only one treatment comparison is of interest in the study and therefore there are no requirements to adjust for multiple comparisons or multiple endpoints within this study.

10.Data Handling Conventions

10.1 Premature Withdrawal and Missing Data

If any subject withdraws prematurely from the study (prior to the final visit), they are

required to complete the withdrawal visit in the CRF. The reasons for withdrawal will be presented in a summary table. For the purposes of summaries and analysis of clinic visit data, this visit will be assigned to the next scheduled clinic visit for that subject, regardless of whether the date falls within the next visit window.

Subjects who withdraw before the end of the study, but who do provide at least one post-baseline measure for a particular endpoint, will be included in the analysis.

Subjects who do not attend any visits after randomization will be excluded from analysis of any endpoint, as no post-baseline data will be available.

10.2 Event Rates

The number of people and person-time of events should be recorded in detail and showing the event rate of each treatment group in summary statement.

The event rate for each treatment group will be calculated as: the sum of number of event for all the patients / the sum of number of treatment periods for all the patients.

10.3 Time to Event Analysis

Differences between treatments in the risk of any stroke (ischemic or haemorrhagic) event or combined vascular events during follow-up were assessed using standard Kaplan-Meier time-to-event approaches. The time to the first event was used in the model when there were multiple events of the same type. Patients were considered censored at the time of study termination or death if there were no events occurred during the study.

11.Study Population

2709 11.1 Disposition of Subjects

2710 The number of subjects in each analysis population will be presented, subjects to be
2711 excluded from the Per Protocol population will be listed, and the total number of subjects
2712 attending each clinic visit will also be summarized by treatment group.

2713 The number of subjects randomized, completed and prematurely withdrawn from the
2714 study will be presented for each treatment group. The primary reasons for withdrawal
2715 both prior to and post randomization will also be presented.

2716 A data display listing and summary of deviations from the inclusion/exclusion criteria
2717 will be presented for all subjects who were either entered or randomized into the trial.

2718 11.2 Protocol Deviations

2719 Subject data will be examined for evidence of protocol violators in order to assess how
2720 well the protocol was followed. Inclusion and exclusion criteria are detailed in the study
2721 protocol.

2722 Subjects who commit protocol violations will be included in the FAS (mITT) Population
2723 but excluded from the Per Protocol Population. These protocol violations will be shown
2724 in a listing. Subjects can either be full or partial protocol violators. A full protocol
2725 violator is completely excluded from the Per Protocol Population. A partial protocol
2726 violator has only some data excluded. For subjects who violated the protocol during the
2727 treatment period due to unpermitted changes in the medication or prohibited concurrent
2728 medication, the analysis will only use data recorded prior to the violation. For all
2729 violations which reference the treatment period, the treatment start date will be used as
2730 the reference date.

A listing of all possible protocol violators will be produced for clinical review. The final list of subjects who are protocol violators and are therefore excluded from the Per-Protocol population will be agreed by the study team.

12. Statistical Analysis

12.1 Demographic and Baseline Characteristics

The following demographic information will be listed and summarized for subjects in each treatment group: age, sex, height, weight, vital signs. This will also be done for medical history, medication history, stroke and TIA history.

The continuous data will be presented as median and interquartile range; categorical data will be presented as n(%). Wilcoxon rank sum test will be used for comparison between two continuous data, and Chi-squared tests or Fisher exact test will be used for comparison between two categorical data.

12.2 Primary Efficacy Endpoint

The composite event rates and corresponding 95% CIs of the primary outcome in the two treatment groups will be estimated by Kaplan-Meier survival analysis and compared by log-rank test. Hazard ratio between the two groups and its 95% CI will be calculated using a Cox proportional hazards regression model.

The main analysis of this study will be conducted based on intention-to-treat principle, and a per protocol analysis will also be conducted as sensitivity analysis. ITT, PPS and ATS will be used to analyze for the primary efficacy outcome, respectively. If the results in the PPS and ATS population are inconsistent with the ITT population,

2753 detailed analysis of the inconsistent results is required.

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2755 **Subgroup Analysis**

2756 Summary tables will be produced for the predefined subgroups and interactions between
2757 treatment and these subgroups will be investigated, using a Cox proportional hazards
2758 model. A separate model will be used for each interaction to determine its significance.
2759 This will also be presented graphically on a forest plot. Treatment effect modification on
2760 the primary endpoint will be explored in the following subgroups:

- 2761 ✓ Age, <65 years old vs. ≥ 65 years old
- 2762 ✓ Sex, men vs. women
- 2763 ✓ Hypertension, yes vs. no
- 2764 ✓ Diabetes, yes vs. no
- 2765 ✓ Smoking, yes vs. no
- 2766 ✓ Baseline renal function, $\text{eGFR} < 60 \text{ ml/min/1.73m}^2$, $\text{eGFR} \geq 60 \text{ ml/min/1.73m}^2$
- 2767 ✓ Target vessel stenosis degree, $< 80\%$, $\geq 80\%$
- 2768 ✓ BMI, $< 25 \text{ Kg/m}^2$, $25\text{-}30 \text{ Kg/m}^2$, $\geq 30 \text{ Kg/m}^2$
- 2769 ✓ Hypoperfusion, CTP, ASITN/SIR collateral circulation scale
- 2770 ✓ Anterior circulation vs. posterior circulation
- 2771 ✓ The mechanism, ischemic stroke vs. TIA

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2773 **12.3 Secondary Efficacy Endpoint**

- 2774 ✓ Any stroke (ischemic or hemorrhage stroke) or all-cause deaths within 30 days

2775 after enrollment or after balloon angioplasty procedure of the qualifying lesion
2776 during follow-up will be analyzed using a Cox proportional hazard model and
2777 report the hazard ratios with 95% CIs;

2778 ✓ Any stroke (ischemic or hemorrhage stroke) inside of the territory of the target
2779 artery or all-cause deaths (within 90 days, 12 months, 24 months and 36 months
2780 after enrollment) will be analyzed using a Cox proportional hazard model. The
2781 hazard ratios with 95% CIs will be reported;

2782 ✓ Any stroke (ischemic or hemorrhage stroke) outside of the territory of the target
2783 artery (within 90 days, 12 months, 24 months and 36 months after enrollment)
2784 will be analyzed using a Cox proportional hazard model. The hazard ratios with
2785 95% CIs will be reported;

2786 ✓ Combined events such as stroke, myocardial infarction, and vascular death
2787 (within 12 months, 24 months and 36 months after enrollment) will be analyzed
2788 using a Cox proportional hazard model. The hazard ratios with 95% CIs will be
2789 reported.

2790 ✓ Common odds ratios of mRS (at 90 days, 12 months, 24 months and 36 months)
2791 will be estimated using ordinal logistic regression.

2792 ✓ Logistic regression analysis will be performed to compare the restenosis rate of
2793 the target artery within 12 months (defined as rate of stenosis >70% or increased
2794 by 30% based on following-up neurovascular imaging) between the two treatment
2795 groups and the odds ratio with 95% CI will be reported.

2796 ✓ Life quality assessment (EuroQol-5-Dimensions Scale [EQ-5D] questionnaire)
2797 within 12 months after enrollment will be tested using student t-test or Wilcoxon

2798 rank sum test as appropriate.

2799 12.4 Safety Analyses

2800 ***Adverse Events***

2801 Adverse events (AEs) will be coded using the MedDRA coding dictionary (Version 26 or
2802 a later release) and grouped by system organ class. Separate data display listings and
2803 summaries will be presented for adverse events. Within each treatment group, the number
2804 and percentage of subjects experiencing an AE will be summarized by system organ class
2805 and preferred term and Fisher's Exact test will be used to compare the number of each
2806 grouped AE event between treatment groups.

2807 ***Deaths and Serious Adverse Events***

2808 Summary tables and data displays will be provided for serious adverse events (as detailed
2809 in the study protocol). In addition, all deaths and serious AE's will be documented in a
2810 case narrative format in the clinical study report.

2811 The number of deaths occurring over the treatment period will be summarized and
2812 Fisher's Exact test will be used to compare the number of deaths between treatment
2813 groups.

2814 ***Vital Signs***

2815 The results of vital signs at different time points will be statistically described and
2816 compared.

2817 12.5 Concomitant Medication

2818 Describe the list of concomitant medication including antiplatelet drug (sspirin,
2819 clopidogrel, ticagrelor, cilostazol), anticoagulation drug, lipid-lowering drug (statin),

antihypertensive drugs (diuretic, angiotensin-converting enzyme inhibitor, angiotensin receptor blocker, beta-blocker, calcium channel antagonist, central α agonist, vasodilator) and so on. The Chi-squared tests or Fisher exact test will be used to compare the number between treatment groups.

13.Reference

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2871 **Revisions to previous SAP version**

SAP version 1.0	Changes in SAP version 2.0
5.3 Interim Analyses	5.3 Interim Analyses
Interim analysis is performed when half of patients finished 12-months visit. The mid-term analysis α 1 is estimated to be 0.0015, and the final analysis α 2 is 0.024.	Interim analyses will not be performed in this trial. However, a Data and Safety Monitoring Board (DSMB) is in place to ensure the safety of subjects in the study. An independent statistician will prepare summary statistics on enrollment, subject status in the study, baseline characteristics, and safety data, including summary tables of coded SAEs and these will be examined by the DSMB. These tables will be provided to the DSMB at regular intervals. If the tables give rise to safety concerns for any treatment, the DSMB may recommend that the trial should be modified or stopped prematurely. The Steering Committee will, in conjunction with the Sponsor, decide whether to act on this recommendation.
6. Sample Size Estimates	6. Sample Size Estimates
The estimation of the sample size is calculated according to the 1-year results of the SAMMPRIS trial, the incidence of target vessel stroke or death rate was 12.2% within 12 months after enrollment in the experimental group. The sample size was 686 cases, with a 80% power, two-sided α = 0.05, to detect a 50 % relative reduction (6.1%) after balloon angioplasty for intracranial atherosclerosis. It is assumed that the O'Brien-Fleming spending function will lead to 6% expansion of the total	According to VISSIT trial and a previous randomized trial in China, the composite event rate of the primary outcome in the medical group is anticipated to be 15%. As to the balloon angioplasty group, we assume a 7% of the primary outcome based on studies of angioplasty without stenting and investigators's clinical practice experience in China. As a result, the sample size required to detect an absolute difference of 8%. A total of 512 patients (256 per group) will be enrolled

type I error. The SAS software is used to calculate the sample size required for the mid-term analysis, which increases the sample size to the required 728 cases. Considering the 10% lost to follow-up rate, the final sample size of this study is 802 cases.	considering an 80% statistical power at one-sided α of 2.5% and a 10% dropout rate.
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